

Chiral, Custodial, and Infrared Symmetries

Integrated Limit-Ancestry, Vacuum-Realization, and UV-to-IR Occupancy Closure in the
Standard Model

Amos Jay Maley

7 August 2026

Release v3.1.0

AASC Standard Model Symmetry Arc – SYM-6

Release status. Audit-grade integrated manuscript and executable semantic certificate. The fixed target family, explicit faithful chiral quotient, computed discrete-axial kernel, exact three-flavor left-background subgroup, background-completed Skyrmion matching, registered vacuum realization, chiral-EFT transport, common-fiber approximation orders, branch-atomic custodial hierarchy, HQET field construction, canonical normal-form atlas, fingerprinted ancestry category, declared-domain exhaustion, no-promotion theorem, and downstream exports are synchronized in one deterministic project.

Arc module	SYM-6, importing the frozen SYM-0 through SYM-5 interfaces and the post-matrix Standard Model and Higgs carrier arcs.
Endpoint	Integrated effective-symmetry ancestry, normalized occupancy, declared-domain exhaustion, and no-primitive-promotion closure.
Declared inventory	Six families, eighteen atomic records, forty named results, twenty release gates.
Primary exports	Custodial packet to SYM-8; scale/RG/EFT packet to SYM-10; anomaly/IR packet to the all-arc capstone.
Physics boundary	QCD vacuum witness, Wilson coefficients, low-energy constants, hadronic matrix elements, and empirical scale comparisons remain registered conventional inputs where stated.

Abstract

Chiral QCD, the anomalous singlet axial current, custodial structure, isospin, hadronic $SU(3)$, and heavy-quark symmetry are ordinarily presented in different targets and at different levels of exactness. This paper constructs a single target-faithful ancestry and transport system for those sectors without collapsing classical invariance, quantum anomaly, vacuum realization, explicit deformation, approximation, EFT emergence, topology, or renormalization into one status label. The declared domain is refined into eighteen atomic records. Each record receives an eleven-field normal form

$$\text{NF}_6(s) = (T_s, G_s, L_s, J_s, A_s, V_s, \Delta_s, E_s, M_s, O_s, H_s)$$

and a source-identified evidence certificate.

The conventional QFT layer constructs the faithful massless-QCD action and center quotient, nonsinglet and singlet Ward identities, the discrete axial residue, conditional vacuum stabilizer and Goldstone realization, anomaly-matching/Wess-Zumino-Witten transport, exact vector-flavor limits, chiral spurion counting, the scalar/gauge-Higgs/matter custodial hierarchy, and the heavy-field decomposition and $1/m_Q$ expansion underlying HQET. The QCD chiral vacuum witness is registered as a conventional physics input; its existence and numerical value are not derived here. AASC is applied after each physical object and map is fixed: it governs target identity, admissible comparison, reference-preserving transport, certificate fixation, representative equivalence, multi-cause separation, and primitive-role classification.

The capstone proves that every declared candidate is decided, every admitted record has a complete source-to-handoff path, normalization is unique up to invariant-preserving representative change, all eighteen slots are covered, and no effective, approximate, anomalous, or vacuum-realized symmetry becomes a new primitive Standard Model carrier merely through infrared utility. The conclusion is exact over the declared inventory, not over all BSM or generalized symmetry sectors. The release freezes a custodial interface for SYM-8 and an EFT/RG transport interface for SYM-10, with machine-readable records and an executable finite audit.

Results at a glance

Construction	Audit-grade conclusion
Faithful chiral action	The massless target carries the quotient of $SU(N_f)_L \times SU(N_f)_R \times U(1)_V$ by the generated flavor-center and color-center kernel appropriate to the declared physical algebra; the quotient is calculated from the action.
Axial ledger	The ABJ divergence, explicit quark-mass term, discrete axial residue, and background-field 't Hooft anomaly occupy distinct certificate fields and cannot repair or substitute for one another.
Vacuum and IR	Given the registered QCD vacuum witness, its stabilizer is the diagonal vector subgroup; the conditional Goldstone and nonlinear realization follow, and the WZW level transports the UV anomaly.

Construction	Audit-grade conclusion
Approximate flavor	Isospin and hadronic $SU(3)$ are exact only on their decoupled equal-mass targets. Their physical deformations are spurion graded; limit order is derived inside a common ambient fiber, while EFT accuracy is ordered only on one exact target and observable class.
Custodial package	Scalar, gauge-Higgs, and fermion-coupled custodial statements are different targets. Hypercharge, Yukawa splitting, the minimal-SM lepton obstruction, and higher operators are separately typed. $\rho = 1$ and $T = 0$ are diagnostics, not definitions.
HQET package	Heavy spin/flavor structure follows from the fixed-velocity field split and leading action. Kinetic, chromomagnetic, current-matching, and higher-order corrections receive operator-level grades; the heavy-hadron occupancy predicate excludes top.
Integrated closure	Eighteen atomic records, one normalized record per slot, complete typed ancestry, zero undecided declared slots, and no effective-to-primitive carrier promotion.
Executable audit	Closed semantic namespaces, fingerprinted maps, representative canonicalization, two record digests, collision tests, theorem-DAG acyclicity, validated reachability, exact export payloads, mutation failures, and derived release gates are checked by the bundled validator.

Reading rule

Every substantive result follows the same dependency order: fixed target; framework-neutral QFT construction; faithful action or quotient; anomaly/vacuum/deformation/EFT status; invariant observable or matching certificate; then AASC role and rival disposition. The structural layer never manufactures a field, condensate, coefficient, or QFT identity missing from the conventional construction.

Exact claim ceiling

This paper does not claim an unconditional first-principles proof of $\langle \bar{q}q \rangle \neq 0$, numerical f_π or chiral low-energy constants, a hadron spectrum, a finite-temperature QCD phase diagram, numerical custodial-breaking corrections, a precision-electroweak fit, numerical HQET Wilson coefficients or form factors, $N_f = 3$ from generation cardinality, anomaly removal by mass or vacuum choice, universal BSM symmetry exhaustion, or an exact heavy-quark symmetry outside the stated leading EFT envelope. These are theorem-level exclusions, not editorial qualifications.

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Part I

Fixed domain, architecture, and proof contract

1 The missing effective-symmetry layer

1.1 Problem and endpoint

The fixed Standard Model carrier is not the same object as every symmetry that becomes useful after a limit, a vacuum choice, a decoupling step, or an EFT reduction. Massless QCD has a chiral flavor action that is not the weak-chiral gauge carrier. The singlet axial transformation has a classical current whose quantum divergence is obstructed. Isospin and hadronic $SU(3)$ become exact only on declared equal-mass, QED-off strong-sector targets. Custodial symmetry is a nested scalar, gauge-Higgs, and matter statement rather than an additional gauge factor. Heavy spin and flavor symmetry arise only after the fixed-velocity heavy-mode reduction. These facts are standard separately; their target changes and status coordinates are rarely closed as one ancestry system.

The endpoint of this paper is the finite declared family

$$\mathcal{R}_6 = \{\chi\text{QCD}, U(1)_A, SU(2)_{\text{iso}}, SU(3)_{\text{had}}, SU(2)_{\text{cust}}, HQ_{N_h}\},$$

refined into the eighteen atomic records of Table 2. The endpoint is not a list. For every atomic candidate s the paper constructs a typed source-to-handoff path and the normalized record

$$\text{NF}_6(s) = (T_s, G_s, L_s, J_s, A_s, V_s, \Delta_s, E_s, M_s, O_s, H_s). \quad (1)$$

The terminal claims are existence, normalized uniqueness, multi-cause compatibility, declared-slot coverage, and no primitive-carrier promotion.

1.2 Dependency and export graph

The proof order is fixed:

$$\begin{aligned} \text{target} &\longrightarrow \text{QFT object} \longrightarrow \text{faithful action} \longrightarrow \text{status vector} \\ &\longrightarrow \text{matching/observable} \longrightarrow \text{AASC role} \longrightarrow \text{certificate closure.} \end{aligned}$$

This order blocks two authority reversals. AASC cannot be used to manufacture a missing QFT existence fact; conversely, the availability of an effective QFT symmetry cannot by itself confer primitive carrier standing.

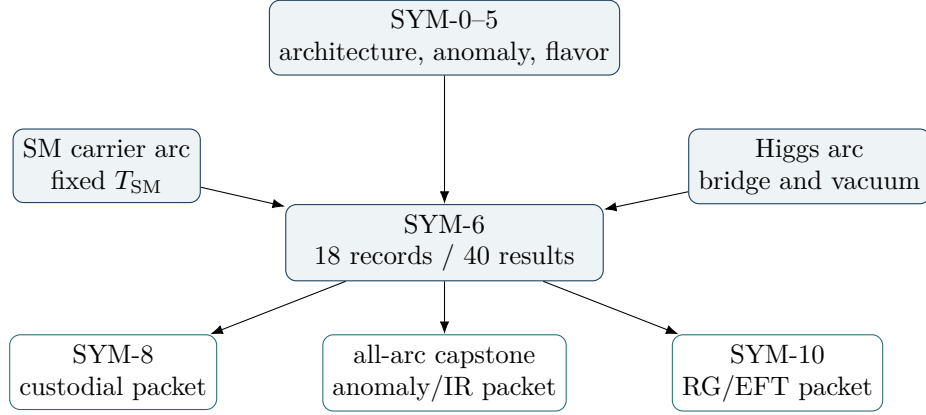


Figure 1: Theorem-level dependency and export graph. The arrows carry registered packets, not title-level resemblance.

1.3 Atomic inventory

Table 2: Declared atomic inventory.

Family	Stable object ID	Required distinction
χ QCD	CHI-NONSINGLET	faithful nonsinglet left/right action and quotient
χ QCD	CHI-VECTOR	diagonal/vector action and baryon-number normalization
χ QCD	CHI-VAC-DIAG	registered vacuum stabilizer and nonlinear realization
axial	AX-CONT	continuous classical action and ABJ obstruction
axial	AX-DISC	faithfully acting discrete axial residue
axial	AX-THOFT	background-field anomaly and IR matching obligation
isospin	ISO-STRONG	exact two-flavor vector limit
isospin	ISO-PHYS	mass/QED-deformed physical record
hadronic $SU(3)$	HAD3-STRONG	exact equal-light-mass vector limit
hadronic $SU(3)$	HAD3-PHYS	strange-light/QED-deformed record
custodial	CUST-SCALAR	scalar-potential $SU(2)_L \times SU(2)_R$ action
custodial	CUST-GH	$g' \rightarrow 0$ gauge-Higgs limit and diagonal subgroup
custodial	CUST-MATTER-MIN	minimal-SM fermion/Yukawa completion and lepton obstruction
custodial	CUST-MATTER-DIRAC	Dirac-neutrino fermion/Yukawa completion
heavy quark	HQ-SPIN	leading fixed-velocity heavy-spin action
heavy quark	HQ-FLAVOR	common-velocity heavy-flavor action
heavy quark	HQ-SPINFLAVOR	combined leading spin-flavor record
heavy quark	HQ-CORR	matching and $1/m_Q$ correction hierarchy

1.4 Formal endpoint statement

The terminal theorem in Section 25 proves the following conditional in its exact finite scope. If the registered upstream propositions and physical existence witnesses hold, and if the sector constructions in Sections 6–19 validate, then every declared atomic slot has exactly one normalized record up to representative equivalence; each record has a certified ancestry path; all simultaneous status causes remain separately recoverable; and no record is a residual primitive Standard Model carrier. Any same-target counterexample to a sector calculation, a missing declared candidate, an invalid import specialization, a graph or normalization failure, or a kernel-faithful primitive-role survivor defeats the endpoint.

2 Fixed targets, maps, and exact nonclaims

2.1 Target fingerprints

A target is not just a Lagrangian name. It is the tuple

$$\tau(T) = (\text{fields, local gauge action, global form, couplings, backgrounds,} \\ \text{state/vacuum, scale window, operator algebra, branch}).$$

Two statements with different non-skin entries of τ are statements on different targets. This is what makes “turn off QED”, “integrate out the weak sector”, and “take $m_Q \rightarrow \infty$ at fixed v ” mathematical operations rather than prose conveniences.

Table 3: Frozen target family.

Target ID	Required content	Principal use
T_{SM}	fixed gauge, matter, Higgs, Yukawa, anomaly, vacuum, and neutrino branch	UV architectural anchor
$T_{\text{QCD}}^{(N_f)}$	$SU(N_c)$ gauge target, declared active flavors, M , θ , optional sources; QED/EW toggles	exact chiral/axial construction
$T_{\chi\text{EFT}}^{(N_f)}$	coset field, derivative expansion, matching packet, scale and scheme	IR nonlinear realization
$T_{\text{had}}^{(2)}$	two-flavor strong sector after EW decoupling; QED toggle	isospin
$T_{\text{had}}^{(3)}$	three-flavor strong sector after EW decoupling; QED toggle	vector hadronic $SU(3)$

Target ID	Required content	Principal use
$T_{\text{cust}}^{(0)}$	one-doublet scalar kinetic/potential target	scalar custodial action
$T_{\text{cust}}^{(1)}$	gauge-Higgs target with g' coordinate explicit	gauge-Higgs custodial limit
$T_{\text{cust}}^{(2,\text{min})}$	fermion-coupled minimal-SM target without ν_R	branch-qualified matter result
$T_{\text{cust}}^{(2,D)}$	fermion-coupled Dirac-neutrino target with ν_R	full doublet alignment result
$T_{\text{HQET}}^{(N_h,v)}$	declared heavy species, common velocity sector, μ , scheme, $1/m_Q$ order	heavy spin/flavor emergence

Let $\mathfrak{G}_6$ be the registered directed multigraph. Its canonical generator types are restriction, decoupling, vacuum realization, faithful quotient, EFT matching, RG transport, representative equivalence, construction, anomaly specialization, observable extraction, normalization, export, and registered assumption. Each generator is the typed tuple

$$e = (id, src, \tau_{src}, dst, \tau_{dst}, type, \mu, w, p, q),$$

where μ is an exact registered map, w a theorem or registered-input warrant, p a set of preserved-data IDs, and q a set of registered assumptions. Only representative-equivalence generators may leave every invariant normal-form coordinate unchanged.

Theorem 2.1 (SYM6-F01: fixed target-family formation). *The path category*

$$\mathbf{C}_6 = \text{FreeCat}(\mathfrak{G}_6)$$

has the registered targets, records, observables, diagnostics, and packets as objects; certified ledger generators as generating morphisms; empty paths as identities; and endpoint-matched path concatenation as composition. No untyped target switch is an admissible morphism.

Proof. Each object is fixed by a registry fingerprint. The free-category construction supplies an identity at every object and associative composition by concatenation of finite paths only when both the node ID and endpoint fingerprint agree. The map registry independently fixes the same source, target, fingerprints, type, warrant, preserved-data set, and assumption set; any disagreement rejects the generator before reachability is computed. Every result warrant declares source-fingerprint and target-fingerprint scopes, a record-target-class scope, a family scope, and admitted edge types. A generator is accepted only when both endpoints and its type lie in those scopes; when its destination is a record, the destination's (NF.T) fingerprint and target class must also agree. Physical validity

resides in the theorem or registered input attached to the generator: free-category formation organizes certified arrows and does not manufacture a physical arrow from syntax. $\square$

2.2 Forbidden silent switches

The following substitutions are invalid unless a named map has been proved: $N_f = 3$ from three generations; an equal-mass vector limit for the massless chiral target; a QED-off hadronic target for the full Standard Model; the scalar custodial target for the fermion-coupled target; the fixed-velocity leading HQET target for finite-mass QCD; a three-flavor WZW certificate for arbitrary N_f ; or a basis change for removal of a physical anomaly. These are structural type errors before they are physics errors.

2.3 Exact nonclaims

The claim ceiling printed before the contents is binding. In particular, the registered vacuum statement in Section 8 is a conditional source of a stabilizer and Goldstone construction, not a dynamical derivation of a condensate. The EFT constructions expose coefficient slots; they do not fill them numerically from the AASC kernel. The no-promotion theorem concerns primitive role, not physical existence or usefulness.

3 Mainstream QFT ledger and source authority

3.1 Construction ledger

Table 4: Physics authority ledger.

Sector	Physical object or proposition	Status here	Primary authority
massless QCD	left/right action, currents, mass spurion	CONSTRUCTED HERE	direct variation; Gasser and Leutwyler [1]
global form	faithful flavor/baryon quotient	CONSTRUCTED HERE	action-kernel calculation; Tanizaki [2]
singlet axial	regulated divergence and anomaly class	specialized import	Adler [3], Bell and Jackiw [4]; SYM-3
QCD vacuum	conventional chiral witness	REGISTERED INPUT	source-identified assumption; lattice/phenomenological status not recomputed
Goldstone and ChPT	conditional coset and EFT	CONSTRUCTED HERE	Gasser and Leutwyler [1, 5], Ecker [6]

Sector	Physical object or proposition	Status here	Primary authority
WZW	quantized anomaly transport	specialized import/new match	Tanizaki [2], Wess and Zumino [7], Witten [8]; SYM-3
isospin/ $SU(3)$	exact vector limits and deformations	CONSTRUCTED HERE	direct stabilizer; ChPT sources
custodial	bidoublet, nested limits, diagnostics	CONSTRUCTED HERE	Sikivie et al. [9], Kribs et al. [10] and Higgs arc
HQET	mode split, leading action, corrections	CONSTRUCTED HERE	Georgi [11], Eichten and Hill [12], Mannel et al. [13], Neubert [14], Cohen et al. [15]
top boundary	decay-before-hadronization ordering	REGISTERED INPUT	conventional lifetime/scale input

3.2 Imported SYM and structural packets

The import identifiers, versions, assumptions, consumers, and limitations are frozen in `sym6_imports.csv`. Four SYM-3 packets are named because their specialization boundary is load-bearing:

- **ANOM-F-SM-LOCAL-1GEN**: aggregate one-generation local gauge ledger; used only as UV consistency context;
- **ANOM-F-VA-BARDEEN-DESCENT**: vector/axial descent and counterterm redistribution; specialized in A01;
- **ANOM-F-QCD-U1A-NF3**: $N_f = 3$ singlet-axial packet; used only when the target is explicitly three-flavor;
- **ANOM-F-QCD3-WZW-MATCH**: three-flavor WZW packet; transported only to the matching target with the same level and background data.

SYM-5 contributes basis/spurion/orbit-stabilizer discipline but not the QCD chiral group itself. The Standard Model carrier arc fixes the UV carrier target and its no-primitive-escape boundary. The Higgs arc fixes the bridge carrier, vacuum branch, and carrier identity used by the custodial construction. The RG transport paper governs scale, scheme, threshold, and quotient preservation. No imported title is treated as a substitute for a theorem statement.

3.3 Import diff rule

For each import, SYM-6 records one of five dispositions: unchanged use, narrower specialization, transport through a proved map, strengthening by a new construction, or provenance only. General- N_f claims are never certified solely by an $N_f = 3$ packet. A downstream conclusion inherits every limitation

of its imported premise unless a named SYM-6 theorem discharges that limitation.

4 AASC dependency import and kernel applicability

4.1 Kernel status

The AASC kernel consists of Admissibility, Standing, Reference, and Irreversibility/fixation. In this manuscript it is invoked only for already constructed determinate objects: target fingerprints, actions, quotient classes, currents, vacuum witnesses, EFT maps, observables, records, and release certificates. A claim that two descriptions concern the same target already presupposes a fixed reference; a transport claim presupposes an admissible map and standing-bearing source and target; a released result presupposes irreversible fixation. Thus the kernel is antecedent to non-degenerate use, not an optional interpretive decoration.

The dependency lock is

$$\text{NonDegDet}(x) \implies (\text{Adm}(x), \text{St}(x), \text{Ref}(x), \text{Irr}(x)).$$

This implication does not provide the QCD action, anomaly coefficient, Higgs bidoublet, or HQET expansion. Those are physics constructions. It governs whether the resulting objects may be silently changed, repaired by a new representative, double counted, or promoted to a role they do not occupy.

4.2 Fixed-domain identity and no-repair rules

A representative modification is lawful only if it preserves the target, physical action, anomaly class, vacuum/boundary status, EFT envelope, invariant observables, and handoff data. A Bardeen counterterm can redistribute a consistent anomaly between currents but cannot erase a fixed obstruction while preserving all target data. A spurion transformation can expose formal covariance but cannot convert a fixed numerical mass tensor into an unbroken symmetry. A different EFT basis can move coefficient labels but cannot remove an invariant custodial-violating amplitude. These are applications of reference preservation to familiar QFT distinctions.

4.3 Carrier versus effective record

The upstream carrier closure fixes the primitive package required for same-scope Standard Model standing. SYM-6 records act on, organize, or transport states and observables within limits of that package. Their importance is not in question. The no-promotion burden is more precise: does the effective record supply a new, independently necessary carrier of same-scope Standard Model determinacy, or does it depend on fixed gauge, matter, Higgs, anomaly, flavor, vacuum, and RG interfaces? Sections 22–25 answer record by record.

5 Record grammar, equivalence, and evidence certificates

5.1 Eleven-field normal form and eight-coordinate status

For every atomic record, Eq. (1) means:

T	exact target and branch	G	faithful action, not presentation alone
L	exact limit vector	J	currents, charges, or action data
A	local/'t Hooft/global anomaly status	V	vacuum realization
Δ	typed deformation multiset	E	EFT fields, scale, scheme, order
M	matching/transport map	O	invariant observable/diagnostic
H	downstream packet		

The independent status vector is

$$\sigma_6(s) = (C, Q, V, X, P, E, B, R),$$

for classical invariance, quantum anomaly, vacuum realization, explicit deformation, approximation/power counting, EFT emergence, boundary/topology, and RG/matching.

Theorem 5.1 (SYM6-F02: faithful-action quotient rule). *Let $\alpha : G \rightarrow \text{Aut}(\mathcal{A}_T^{\text{phys}})$ be a candidate action on the declared physical algebra. The physical symmetry represented by G is the induced faithful action of $G/\text{Ker } \alpha$. Two presentations define the same G_s field only when their induced actions are isomorphic on the fixed target and preserve all invariant normal-form coordinates.*

Proof. The first isomorphism theorem gives an injective homomorphism $G/\text{Ker } \alpha \hookrightarrow \text{Aut}(\mathcal{A}_T^{\text{phys}})$ with image $\text{im } \alpha$. Any further quotient removes a nontrivially acting element; any smaller quotient retains a redundant element. The second statement follows because target, anomaly, vacuum, EFT, observable, and export data are part of the identity predicate $\sim_6$ below. Thus Lie-algebra equality or group dimension alone is insufficient. $\square$

Definition 5.1 (Representative equivalence). For admissible records r, r' , write $r \sim_6 r'$ exactly when they have the same target fingerprint, isomorphic faithful action on the physical algebra, identical anomaly and vacuum/boundary data, the same EFT and matching envelope, identical invariant observables, and identical handoff payload. Basis, current-improvement, quotient-presentation, or operator-basis changes are skin changes only when these invariants are preserved.

Theorem 5.2 (SYM6-F03: orthogonal status separation). *No coordinate of σ_6 is a function of any one other coordinate over the declared target family. In particular, anomaly, vacuum realization, explicit deformation, approximation, and EFT emergence cannot substitute for one another in a complete record.*

Proof. The declared records supply separating witnesses. Nonsinglet massless-QCD currents are classically and quantum conserved while the registered vacuum realizes them nonlinearly. The singlet axial current is classically conserved at $M = 0$ but anomalous quantum mechanically, independently of a vacuum. Isospin is nonanomalous but explicitly deformed when $m_u \neq m_d$ or QED is on. The scalar custodial action is exact on $T_{\text{cust}}^{(0)}$ while not an exact full-matter action. Heavy spin is an EFT-emergent leading invariance with controlled $1/m_Q$ violation. These pairs agree in one coordinate while differing in another, so the proposed substitutions fail. The incidence matrix in Section 21 exhausts the declared combinations. $\square$

5.2 Certificate object

Each normal form is accompanied by

$$\text{Cert}_6(s) = (id, p, q, w, d, m, c, x, v),$$

where id is stable identity, p provenance, q target/assumptions, w witnesses, d theorem dependencies, m machine status and matching data, c rival disposition, x exact nonclaims, and v release version/hash. Required fields may be marked not applicable only by a type rule. “Unknown”, a blank string, or an untyped null fails release.

Theorem 5.3 (SYM6-F04: normal-form admissibility). *A candidate record is admitted if and only if all type-required fields of NF_6 and Cert_6 are typed, registry-resolved, mutually compatible, target-faithful, evidence-bearing, and semantically valid. Every ID-bearing path has exactly one resolver namespace. Provenance, imported packets, results, and assumptions carry target-class, family, and result or edge-type scopes and must be compatible with the consuming record. The full representative receives a content digest, while its canonical projection receives a distinct invariant digest after presentations, current improvements, operator bases, and registered scheme transports are replaced by their invariant classes.*

Proof. Necessity is the conjunction defining an evidence-bearing record. For sufficiency, each sector theorem supplies the physical proposition or witness, while the closed schema and registries decide the finite serialization obligations: exact target/branch compatibility; current domain; anomaly flavor/background scope; vacuum-witness branch; EFT envelope, order, operator basis, and scheme class; matching endpoints; provenance/import scope; result scope; assumption scope; observable; and handoff. The import table and provenance registry must agree on their declared scopes, and every consuming record must lie in the relevant intersections. Unknown namespaces and incompatible registered IDs fail independently of a recomputed digest. Canonical uniqueness is proved globally in Theorem 21.2; F04 supplies semantic admission and never substitutes datatype validation for sector evidence. $\square$

5.3 Representative versus invariant ledger

Table 5: Representative and invariant distinctions.

Change	Normally representative	Becomes a new record when it changes
quark flavor basis	matrix entries, chart phases	orbit invariants or stabilizer stratum
current improvement	local current representative	charge/action domain or anomaly class
counterterm convention	anomaly distribution among currents	gauge invariance or fixed physical obstruction
coset chart	coordinates of $U(x)$	target, global quotient, WZW level
EFT operator basis	coefficient labels	invariant amplitude, truncation, matching order
HQET velocity label	reparameterization within overlap	velocity superselection target or domain
custodial notation	bidoublet convention	acting group, branch, spurion content

The record grammar is now fixed. Sector work that follows either populates these fields or fails visibly; no later chapter may weaken the schema to accommodate a missing construction.

Part II

Chiral QCD, axial obstruction, and vacuum realization

6 The massless QCD target and faithful global group

6.1 Target action and active-flavor rule

Fix $N_c \geq 2$ and a declared set F of $N_f \geq 2$ active Dirac flavors. “Active” is a target choice: their masses and the scale window are included explicitly. It is not generation number and is not inferred from kinematic production alone. With electroweak interactions and QED decoupled, optional flavor backgrounds l_μ, r_μ , mass source M , and vacuum angle θ , write

$$\begin{aligned}
\mathcal{L}_{\text{QCD}}^{(N_f)} = & -\frac{1}{4}G_{\mu\nu}^a G^{a\mu\nu} + \bar{q}_L i \not{D} q_L + \bar{q}_R i \not{D} q_R - \bar{q}_L M q_R - \bar{q}_R M^\dagger q_L \\
& + \frac{\theta g_s^2}{32\pi^2} G_{\mu\nu}^a \tilde{G}^{a\mu\nu},
\end{aligned} \tag{2}$$

where D may include l_μ, r_μ as nondynamical sources. At $M = 0$ the presentation action is

$$q_L \mapsto e^{i\beta} L q_L, \quad q_R \mapsto e^{i\beta} R q_R, \quad L, R \in SU(N_f), \quad e^{i\beta} \in U(1)_q.$$

The label $U(1)_q$ gives a quark charge of one; the baryon-normalized circle is obtained after the color-center identification.

Theorem 6.1 (SYM6-Q01: massless-QCD target and transformation). *On $T_{\text{QCD}}^{(N_f)}$ with $M = 0$ and QED/EW off, the kinetic and gluon terms in Eq. (2) are invariant under $SU(N_f)_L \times SU(N_f)_R \times U(1)_q$. The transformation extends covariantly to external sources by*

$$l_\mu \mapsto L l_\mu L^\dagger + i L \partial_\mu L^\dagger, \quad r_\mu \mapsto R r_\mu R^\dagger + i R \partial_\mu R^\dagger, \quad M \mapsto L M R^\dagger.$$

Proof. Unitarity cancels L, R in the flavor contractions and both commute with the color covariant derivative. The source transformations make $D_\mu q_{L,R}$ transform like $q_{L,R}$. The mass term is covariant under the stated bifundamental rule and is invariant when M is transformed as a source; a fixed M retains only its stabilizer. No use has been made of generation cardinality. $\square$

6.2 Kernel of the presentation action

Let $\omega_f = e^{2\pi i/N_f}$ and $\omega_c = e^{2\pi i/N_c}$. On quarks the common flavor/vector-center element

$$\kappa_f = (\omega_f \mathbf{1}, \omega_f \mathbf{1}, \omega_f^{-1})$$

acts trivially. On gauge-invariant operators, the vector rotation

$$\kappa_c = (\mathbf{1}, \mathbf{1}, \omega_c)$$

is equivalent to a color-center gauge transformation and also acts trivially. If $\kappa_f^a = \kappa_c^b$ in the displayed product, the left flavor component forces $\omega_f^a = 1$ and the vector component then forces $\omega_c^b = 1$. Thus the two subgroups intersect trivially and

$$\Gamma_\chi = \langle \kappa_f \rangle \times \langle \kappa_c \rangle \simeq \mathbb{Z}_{N_f} \times \mathbb{Z}_{N_c}.$$

The quotient data are therefore explicit and do not depend on a common-divisor guess.

Theorem 6.2 (SYM6-Q02: faithful chiral-group classification). *On the gauge-invariant graded local algebra generated by meson operators, N_c -quark baryon operators, currents, and the declared background generating functional, the complete faithfully acting massless-QCD symmetry is*

$$G_{\text{QCD}} = \frac{SU(N_f)_L \times SU(N_f)_R \times U(1)_V \times (\mathbb{Z}_{2N_f})_A}{\mathbb{Z}_{N_c} \times (\mathbb{Z}_{N_f})_L \times (\mathbb{Z}_{N_f})_R \times \mathbb{Z}_2} \simeq \frac{SU(N_f)_L \times SU(N_f)_R \times U(1)_V}{\mathbb{Z}_{N_c} \times (\mathbb{Z}_{N_f})_V}. \quad (3)$$

After defining $U(1)_B = U(1)_V / \mathbb{Z}_{N_c}$ so that an N_c -quark baryon has baryon charge one, this is equiva-

lently

$$G_{\text{QCD}} \simeq \frac{SU(N_f)_L \times SU(N_f)_R \times U(1)_B}{(\mathbb{Z}_{N_f})_V},$$

where $(\mathbb{Z}_{N_f})_V$ is the explicit diagonal vector-flavor/baryon identification. The discrete axial presentation is embedded in this connected quotient rather than appended as an independent factor.

Proof. Writing $U(N_f)_{L,R} = [SU(N_f)_{L,R} \times U(1)_{L,R}]/\mathbb{Z}_{N_f}$ supplies the two flavor-center identifications, while $U(1)_L \times U(1)_R = [U(1)_V \times U(1)_A]/\mathbb{Z}_2$ identifies the simultaneous vector and axial π rotations. The fermion measure reduces $U(1)_A$ to $(\mathbb{Z}_{2N_f})_A$, and the color-center gauge equivalence supplies $\mathbb{Z}_{N_c}$. The axial generator can be absorbed into a right flavor center and a vector phase, giving the second presentation. Quotienting $U(1)_V$ by the color center gives baryon normalization and leaves one diagonal $\mathbb{Z}_{N_f}$ identification. Conversely, a class outside these denominators changes a meson, a baryon, or the background generating functional. Theorem 5.1 then yields the faithful action. This is the explicit global form of Tanizaki [2]. $\square$

6.3 Currents and domains

With T^a Hermitian and traceless, $\text{Tr}(T^a T^b) = \delta^{ab}/2$, define

$$J_V^{a\mu} = \bar{q}\gamma^\mu T^a q, \quad J_A^{a\mu} = \bar{q}\gamma^\mu \gamma_5 T^a q, \quad J_B^\mu = \frac{1}{N_c} \bar{q}\gamma^\mu q.$$

The factor $1/N_c$ implements the baryon-normalized circle. Charges exist on domains where spatial flux terms and boundary conditions make the integral well defined. Local current expressions without that domain are not by themselves a global action certificate.

Theorem 6.3 (SYM6-Q03: current and Ward-identity theorem). *On the source-coupled target, variation of the generating functional yields the nonsinglet Ward identities. With fixed mass source and QED off,*

$$\partial_\mu J_V^{a\mu} = i\bar{q}[M, T^a]q, \tag{4}$$

$$\partial_\mu J_A^{a\mu} = i\bar{q}\{M, T^a\}\gamma_5 q, \tag{5}$$

up to contact terms fixed by insertions. At $M = 0$ both nonsinglet families are locally conserved; baryon number is vectorlike and nonanomalous in QCD.

Proof. Promote the flavor parameter to a compactly supported function and vary Eq. (2). Integration by parts gives the current divergence; variation of the mass term gives the commutator or anticommutator. The traceless generators have no QCD singlet triangle coefficient, while the vector measure is invariant. Source differentiation supplies the contact terms. Boundary and state conditions determine whether the local currents integrate to charges. $\square$

6.4 Mass and electromagnetic spurions

The mass source is bifundamental. When frozen at M_0 , the exact group is the stabilizer

$$\text{Stab}(M_0) = \{(L, R) : LM_0R^\dagger = M_0\}.$$

For the vector subgroup $L = R = V$ this reduces to the commutant $[V, M_0] = 0$. Electromagnetism introduces the charge matrix Q and e , with formal source rules $Q_{L,R} \mapsto LQ_LL^\dagger, RQ_RR^\dagger$ before setting $Q_L = Q_R = Q$. A fixed electromagnetic interaction retains the common commutant of M_0 and Q .

Theorem 6.4 (SYM6-Q04: mass and electromagnetic deformation). *Reintroducing M and QED produces two independent deformation coordinates Δ_M and Δ_{em} . Their exact residual group is the faithful image of the simultaneous stabilizer of the fixed spurions. Neither coordinate changes the pre-existing ABJ anomaly class.*

Proof. The stabilizer statement follows from the source transformations and Theorem 5.1. The two sources transform in different representations and may be varied independently, so they occupy distinct coordinates. Anomaly coefficients are determined by the fermion representation and regulator-consistent measure; adding a mass term changes the Ward identity by an explicit operator but does not cancel the measure Jacobian. QED may add its own anomaly term when retained; it does not repair the QCD term. $\square$

7 Continuous axial current and discrete residue

7.1 The regulated singlet identity

Let $J_5^\mu = \bar{q}\gamma^\mu\gamma_5q$. In the convention of Eq. (2), the renormalized identity is

$$\partial_\mu J_5^\mu = 2i\bar{q}M\gamma_5q + N_f \frac{g_s^2}{16\pi^2} G_{\mu\nu}^a \tilde{G}^{a\mu\nu}, \quad (6)$$

with the first term understood as $i\bar{q}(M\gamma_5 + \gamma_5 M^\dagger)q$ for a general complex source. Normalization changes must transport both the current and coefficient.

Theorem 7.1 (SYM6-A01: continuous axial divergence). *Equation (6) is the target-specialized singlet axial Ward identity. At $M = 0$ the classical continuous action exists but its current is not conserved quantum mechanically. A local counterterm, current improvement, vacuum selection, or mass deformation cannot erase this fixed physical disposition while preserving the same vector-gauge target.*

Proof. The Fujikawa Jacobian or triangle computation gives the second term with coefficient fixed by N_f fundamental Dirac fermions [3, 4]. The imported SYM-3 Bardeen packet fixes vector gauge invariance and permits only redistribution among presentation currents. The integrated topological

density changes the effective action under a continuous axial rotation. Any proposed erasure either changes the gauged-current convention, the target, or the anomaly class, so it is not a representative repair. $\square$

7.2 Discrete transformation and theta shift

Under $q_L \mapsto e^{-i\alpha} q_L$, $q_R \mapsto e^{i\alpha} q_R$, the measure shifts $\theta \mapsto \theta - 2N_f \alpha$ in the chosen convention. At fixed θ the measure is invariant for $\alpha = \pi k/N_f$, giving the presentation residue $(\mathbb{Z}_{2N_f})_A$. Let a denote its generator,

$$q_L \mapsto e^{-\pi i/N_f} q_L, \quad q_R \mapsto e^{\pi i/N_f} q_R.$$

In the connected faithful group it is the distinguished class

$$\iota_A(a) = [\mathbf{1}_L, e^{2\pi i/N_f} \mathbf{1}_R, e^{-\pi i/N_f}].$$

Theorem 7.2 (SYM6-A02: discrete axial-residue theorem). *For the declared gauge-invariant graded local algebra $\mathcal{A}_{\text{QCD}}^{\text{gr}}$, the action kernel of the axial presentation is*

$$K_A = \ker\left((\mathbb{Z}_{2N_f})_A \xrightarrow{\iota_A} G_{\text{QCD}} \longrightarrow \text{Aut}(\mathcal{A}_{\text{QCD}}^{\text{gr}})\right) = \begin{cases} \{1\}, & N_c \text{ odd}, \\ \langle a^{N_f} \rangle \simeq \mathbb{Z}_2, & N_c \text{ even}. \end{cases}$$

Hence the faithful image has order $2N_f$ for odd N_c and N_f for even N_c . It is a distinguished subgroup/presentation of G_{QCD} , not an independent direct-product factor.

Proof. The theta periodicity gives the presentation subgroup and the displayed embedding follows by multiplying the right-handed quark by its flavor-center phase and both chiralities by the stated vector phase. The power a^{N_f} sends every quark to its negative, so it is fermion parity. Meson generators contain an even number of quark fields and are unchanged. A baryon generator contains N_c quarks and therefore receives $(-1)^{N_c}$. For odd N_c it detects a^{N_f} , and no smaller nontrivial power acts trivially on both meson and baryon generators. For even N_c , every gauge-invariant local generator has even quark parity, so a^{N_f} is trivial; the quotient by this order-two subgroup is faithful. Changing the operator algebra or retaining fermion parity as independent data requires a new target and a recalculation, not a symbolic K_A . $\square$

7.3 Exact three-flavor background subgroup

The generic result A02 is not itself the background used by the three-flavor WZW certificate. Specialize to $N_f = N_c = 3$ and let $v(\pi/3)$ be the quark-number vector phase acting by $e^{\pi i/3}$ on both chiralities. Inside the full faithful vector-axial global group define

$$\ell = v(\pi/3)a^{-1}. \tag{7}$$

Then

$$\ell : q_L \mapsto e^{2\pi i/3} q_L, \quad q_R \mapsto q_R, \quad \ell^3 = 1.$$

No smaller positive power is the identity, so $\langle \ell \rangle \simeq (\mathbb{Z}_3)_L$. Equation (7) is not a homomorphism extracted from the axial residue alone: it is a composite subgroup relation in the faithful vector–axial quotient. The map registry cryptographically references a certificate that carries the axial generator, compensating vector phase, left/right action, order, complete one- and two-form background data, quotient-compatible ultraviolet and infrared targets, and the A03/M01/M02 evidence witnesses and assumptions. The audit recomputes the certificate digest and requires the map, action registry, background registry, quotient endpoints, and evidence lists to agree exactly.

7.4 Three anomaly coordinates

Local ABJ nonconservation is a statement about a current in a dynamical gauge background. A 't Hooft anomaly is an obstruction to gauging an otherwise meaningful global background action and is invariant under RG flow. A global anomaly is a topology-sensitive phase not represented solely by a local perturbative polynomial. These have different inputs and different consequences.

Theorem 7.3 (SYM6-A03: ABJ/'t Hooft/global anomaly separation). *Every anomaly datum used by SYM-6 has exactly one primary type among local-current, background-gauging, and global/topological, together with any explicitly registered mixed incidence. IR reproduction of a 't Hooft anomaly does not restore the anomalous singlet current; absence of a perturbative polynomial does not prove absence of a global anomaly.*

Proof. The definitions use different test objects: local divergence, background gauge variation, and large/topological transport. The SYM-3 packet and Tanizaki [2] provide examples where a discrete axial/flavor/baryon background anomaly persists although it is not the same object as Eq. (6). In the admitted realization the generic A02 action is specialized to $N_f = N_c = 3$ and composed with the vector phase by Eq. (7); this fixes the exact $(\mathbb{Z}_3)_L$ background used downstream. Since the tests are inequivalent, replacement across columns is invalid. Mixed cases are recorded as relations, not collapsed categories. $\square$

Table 6: Anomaly-type matrix.

Record	Test	Consequence	Cannot imply
ABJ singlet	$\partial_\mu J_5^\mu$	no conserved continuous singlet charge	absence of a discrete residue
't Hooft packet	background variation	IR matching constraint	unique IR phase or restored current
global/topological	large transformation/bordism phase	topology-sensitive obstruction	perturbative local coefficient

8 Registered vacuum realization and Goldstone transport

8.1 Registered witness

Define the renormalized matrix order parameter

$$(W_\chi)^i{}_j = \langle \Omega | \bar{q}_{Rj} q_L^i | \Omega \rangle, \quad W_\chi \mapsto L W_\chi R^\dagger.$$

Registered input 8.1 (Conventional QCD chiral-vacuum witness). On the declared zero-temperature, zero-density confining QCD target in the standard branch, a renormalized witness exists with

$$W_\chi = v_\chi \mathbf{1}_{N_f}, \quad v_\chi \neq 0.$$

Its source, scheme, state, and evidential status are registered. Neither its existence nor the numerical value of v_χ is derived from the AASC kernel or from the finite record validator.

Theorem 8.1 (SYM6-V01: registered QCD vacuum-witness theorem). *Assumption 8.1 defines an admissible, target-specific witness whose transformation is fixed by Q01 and whose use is conditional. All downstream vacuum conclusions carry this premise; no downstream normalization step strengthens it into an unconditional dynamical theorem.*

Proof. The source and target are explicit, and the bifundamental transformation follows from the quark action. The certificate marks the witness as REGISTERED INPUT. The dependency graph makes V01 a premise of V02–M02. The validator checks provenance and limitation fields, so the assumption cannot be silently replaced by “proved here.” $\square$

8.2 Stabilizer and coset

Theorem 8.2 (SYM6-V02: vacuum stabilizer and coset). *Given Assumption 8.1, the stabilizer of $W_\chi = v_\chi \mathbf{1}$ in the nonsinglet chiral action is the diagonal vector subgroup, with the inherited faithful quotient. The physical vacuum orbit is the corresponding quotient of*

$$\frac{SU(N_f)_L \times SU(N_f)_R}{SU(N_f)_V}.$$

Proof. Invariance requires $L W_\chi R^\dagger = W_\chi$. Since $v_\chi \neq 0$, this gives $L = R$. Conversely every common V stabilizes the witness. Passing through the kernel of Theorem 6.2 identifies representatives that act identically. Thus the diagonal subgroup is calculated as a stabilizer, not inserted by convention. $\square$

Theorem 8.3 (SYM6-V03: Goldstone and nonlinear realization). *Assume locality, relativistic QFT, a well-defined conserved nonsinglet charge on the stated domain, and the vacuum witness of V01. The exact massless target has $N_f^2 - 1$ broken nonsinglet generators and therefore the corresponding Goldstone*

modes. A nonlinear representative is $U(x) \in SU(N_f)$ with

$$U \mapsto LUR^\dagger, \quad \mathcal{L}_2 = \frac{f^2}{4} \text{Tr}(D_\mu U D^\mu U^\dagger) + \frac{f^2 B}{2} \text{Tr}(MU^\dagger + UM^\dagger).$$

The constants f, B are EFT inputs to be matched; symmetry does not determine their numerical values.

Proof. The dimension difference is $2(N_f^2 - 1) - (N_f^2 - 1) = N_f^2 - 1$. Under the registered Goldstone hypotheses, each broken generator supplies a massless mode. The coset admits the standard U representative, and the displayed two-derivative and one-mass-source terms are the lowest invariant terms under the source rules. Numerical coefficients are not fixed by the group action [1, 6]. $\square$

9 Anomaly matching, WZW transport, and topological residue

9.1 UV-to-IR matching square

For $N_f \geq 3$, extend U to a five-manifold B_5 whose boundary is spacetime and use the normalized Wess-Zumino functional

$$\Gamma_{\text{WZ}}[U] = -\frac{i}{240\pi^2} \int_{B_5} \text{Tr}(U^{-1}dU)^5, \quad S_{\text{WZW}} = N_c \Gamma_{\text{WZ}} + S_{\text{gauge}}. \quad (8)$$

The gauged completion S_{gauge} is fixed so its variation under flavor backgrounds equals the UV consistent anomaly. Independence of extension modulo 2π requires the integer level. The $N_f = 2$ case is kept as a separate topology-sensitive specialization rather than inferred from the ordinary five-form.

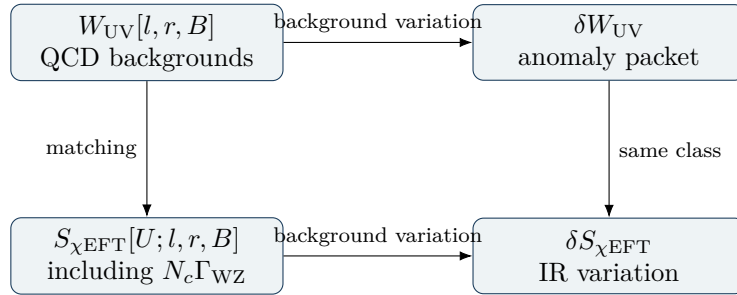


Figure 2: The anomaly-matching square. Commutativity is equality of the registered anomaly class, not equality of microscopic and IR fields.

Theorem 9.1 (SYM6-M01: WZW and anomaly-matching transport). *On the registered three-flavor chiral target, the WZW level $k = N_c$ and its gauged completion reproduce the imported UV non-Abelian flavor anomaly. The matching square in Figure 2 commutes modulo local counterterm convention and $2\pi\mathbb{Z}$ extension ambiguity. For other N_f , a result is admitted only with its own topology and coefficient certificate.*

Proof. The UV anomaly coefficient is proportional to the number of colors because each color contributes

the same flavor triangle. The variation of the gauged WZW functional at integer level k is k times the fundamental flavor anomaly [7, 8]. Setting $k = N_c$ therefore matches the UV variation. Integer quantization makes a change of five-dimensional extension shift the action by $2\pi\mathbb{Z}$. The SYM-3 packet ANOM-F-QCD3-WZW-MATCH fixes the same target and convention. Counterterms alter consistent representatives but not the class. $\square$

9.2 Skyrmion/baryon residue and background completion

For $N_f \geq 2$ the topological current of a smooth U configuration is

$$B_{\text{top}}^\mu = \frac{1}{24\pi^2} \epsilon^{\mu\nu\rho\sigma} \text{Tr}(U^{-1} \partial_\nu U U^{-1} \partial_\rho U U^{-1} \partial_\sigma U). \quad (9)$$

Its integrated charge is the winding number for appropriate boundary conditions. This ungauged current is only the first lemma. Let $\tilde{A}_f$ be the $U(N_f)$ lift of the vector-flavor background with curvature $\tilde{F}_f$, let $A_\chi^{(1)}$ gauge the discrete chiral subgroup, and let $B_f^{(2)}$ be the two-form background implementing the vector-flavor center quotient. For the atomic AX-TH00FT record these are the $N_f = N_c = 3$ backgrounds fixed by Eq. (7). Set

$$U^{-1} DU = U^{-1} [dU + (\tilde{A}_f + A_\chi^{(1)})U - U\tilde{A}_f].$$

The gauge-invariant background completion is the three-form

$$\begin{aligned} J_B[\tilde{A}_f, A_\chi^{(1)}] &= \frac{1}{24\pi^2} \text{Tr}[(U^{-1} DU)^3] \\ &+ \frac{1}{8\pi^2} \text{Tr}[(UDU^{-1})(\tilde{F}_f + dA_\chi^{(1)}) - (U^{-1} DU)\tilde{F}_f]. \end{aligned} \quad (10)$$

The second line is the necessary counterterm completion; naïve covariantization alone is not conserved.

Theorem 9.2 (SYM6-M02: baryon/Skyrmion residue). *Within the ungauged chiral-EFT target, Eq. (9) is identically conserved and carries integer winding. On the registered three-flavor quotient-sensitive background target, Eq. (10) is gauge invariant and obeys*

$$dJ_B[\tilde{A}_f, A_\chi^{(1)}] = \frac{N_f}{(2\pi)^2} dA_\chi^{(1)} \wedge B_f^{(2)}. \quad (11)$$

Consequently the coupling $S_B = \int_{M_4} A_B \wedge J_B$ varies under $A_B \mapsto A_B + d\lambda_B$ as

$$\delta S_B = -\frac{N_f}{(2\pi)^2} \int_{M_4} \lambda_B dA_\chi^{(1)} \wedge B_f^{(2)},$$

which is the boundary variation of

$$-\frac{N_f}{(2\pi)^2} \int_{M_5} A_B \wedge dA_\chi^{(1)} \wedge B_f^{(2)}.$$

This completes the baryon component of the registered discrete axial/ﬂavor anomaly matching. The displayed N_f coefficient records the structural formula; the admitted atomic record and exported packet evaluate it at $N_f = 3$.

Proof. The ungauged current is the pullback of the closed generator of $H^3(SU(N_f), \mathbb{Z})$, so its divergence vanishes for smooth fields and compactified spatial infinity gives the homotopy charge. For the background current, exterior differentiation of Eq. (10), the covariant Maurer–Cartan identity, and the quotient relation between $\text{Tr } \tilde{F}_f$ and $B_f^{(2)}$ give Eq. (11); equivalently it is the descent identity

$$dJ_B = \frac{1}{2!(2\pi)^2} \text{Tr} \left[(\tilde{F}_f + dA_\chi^{(1)})^2 - \tilde{F}_f^2 \right].$$

Integration by parts gives the stated $U(1)_B$ variation and inflow representative, exactly as in Eqs. (4.9)–(4.13) of Tanizaki [2]. The ordinary WZW functional already matches the perturbative flavor anomaly and the appropriate discrete component, but it cannot by itself match this baryon component because $U(1)_B$ acts trivially on U . The completed Skyrmion current supplies the missing response. Ungauged conservation and background anomalous divergence are therefore distinct target statements. No baryon mass, energetic soliton stability, or exhaustive baryon spectrum is derived. $\square$

10 Chiral and axial normal-form closure

10.1 Completed sector records

Table 7: Chiral/axial completion matrix.

Record	Decisive results	Normal-form residue
CHI-NONSINGLET	Q01–Q03	faithful left/right quotient, nonsinglet currents, exact massless limit
CHI-VECTOR	Q02–Q04	diagonal action, baryon normalization, mass-spurion stabilizer
CHI-VAC-DIAG	V01–V03	registered witness, vector stabilizer, coset and Goldstone EFT
AX-CONT	A01	classical action plus ABJ obstruction, not a conserved quantum charge
AX-DISC	A02	explicit axial embedding, parity-sensitive kernel, and faithful image
AX-THOOF	A03, M01–M02	$N_f = N_c = 3$ left-subgroup map, background obstruction, WZW plus completed-Skyrmion matching

Every row has a target, faithful action, exact limit, current/action data, anomaly status, vacuum status, deformation ledger, EFT envelope, matching map, observable, and handoff. For fields that do not apply – for example a Goldstone EFT for the anomalous continuous singlet action as an exact symmetry – the record contains a typed inapplicability reason rather than a blank.

10.2 Rival disposition

The direct-product chiral group fails Q02 because it retains explicit center redundancies. “ $U(1)_A$ is broken” fails F03 because it conflates Eq. (6), mass deformation, the embedded discrete image, and possible vacuum data. “The condensate follows from admissibility” fails V01 because the physical existence witness is registered. “The WZW term follows from the coset” fails M01 because the level and global definition require anomaly input. “The ordinary WZW term alone matches the baryon component” fails M02 because the pion field is baryon neutral; the background-completed current is required. “Skyrmions prove all baryons” exceeds M02. The sector is closed exactly after these rival identifications are excluded, not by omitting them.

Chiral/axial exit certificate

Six of eighteen atomic records are admitted. The faithful quotient and discrete-axial kernel are computed, ABJ/t Hooft/global anomaly types are separated, the QCD vacuum witness remains a declared input, the Goldstone theorem is conditional on its hypotheses, and the background-completed Skyrminion current closes the baryon anomaly component. All six records pass the eleven-field schema and have a typed path to the later ancestry category.

Part III

Isospin, hadronic flavor, and controlled approximation

11 Exact vector-flavor limits

11.1 Two-flavor strong-sector isospin

The target $T_{\text{had}}^{(2)}$ contains gluons and the u, d quarks after weak interactions have been decoupled. Its fingerprint includes the QED toggle and the two-flavor mass matrix

$$M_2 = \hat{m}\mathbf{1}_2 + \frac{m_u - m_d}{2}\tau_3, \quad \hat{m} = \frac{m_u + m_d}{2}.$$

This target differs from the full Standard Model even when its correlators are used to approximate hadronic observables.

Theorem 11.1 (SYM6-P01: strong-sector isospin limit). *On $T_{\text{had}}^{(2)}$ with QED off and $m_u = m_d = \hat{m}$, the action has an exact vector $SU(2)_V$ flavor symmetry, modulo the faithful center identifications inherited from Q02. Its currents are $J_V^{a\mu}$ restricted to $a = 1, 2, 3$. At the physical point the independent leading deformations are $m_d - m_u$ and electromagnetism.*

Proof. The mass matrix is proportional to the identity, so $[V, M_2] = 0$ for every $V \in SU(2)$. The gluon and kinetic terms are flavor blind and the vector current has no QCD anomaly. Turning on $m_d - m_u$ leaves only the commutant of τ_3 ; turning on QED also introduces $Q_2 = \text{diag}(2/3, -1/3)$ with the same diagonal commutant. The statement is exact on the decoupled target, not on unqualified T_{SM} . $\square$

11.2 Three-flavor vector symmetry

Write

$$M_3 = \bar{m}\mathbf{1}_3 + \delta M_8 + \delta M_3,$$

where δM_8 records strange-light splitting and δM_3 is isovector splitting. The physical hierarchy of these terms is empirical; their representation content is structural.

Theorem 11.2 (SYM6-P02: hadronic- $SU(3)$ limit). *On $T_{\text{had}}^{(3)}$ with QED off and $m_u = m_d = m_s = \bar{m}$, the action has exact vector $SU(3)_V$ invariance with the faithful quotient inherited from Q02. Equal nonzero masses do not restore $SU(3)_L \times SU(3)_R$; that enhancement requires $M_3 = 0$.*

Proof. Every $V \in SU(3)$ commutes with $\bar{m}\mathbf{1}_3$, giving vector invariance. A left/right pair with $L \neq R$ transforms the fixed nonzero mass as $LM_3R^\dagger = \bar{m}LR^\dagger \neq M_3$ in general, so only $L = R$ survives. QED introduces a nontrivial charge spurion and reduces the stabilizer further. $\square$

11.3 Limit lattice

Let a limit vector list coordinates

$$\lambda = (N_f, M, e, g_{\text{weak}}, \mu, \text{operator algebra}).$$

The diagram in Figure 3 records restrictions with fixed target maps. A downward arrow sets additional deformation coordinates to zero; a sideways arrow changes the active-flavor target and therefore requires a threshold/decoupling certificate.

Theorem 11.3 (SYM6-P03: flavor-limit refinement). *The chiral, vector- $SU(3)$, isospin, and physical records form the typed lattice of Figure 3. Within one target, setting $M = 0$ refines an equal-mass vector limit to chiral symmetry. Changing N_f is a threshold map, not an equality or a consequence of Standard Model generation cardinality.*

Proof. P01 and P02 compute the vector stabilizers; Q01 computes the massless enhancement. The operations $\delta M \rightarrow 0$ and $\bar{m} \rightarrow 0$ have different stabilizers and therefore different records. Removing

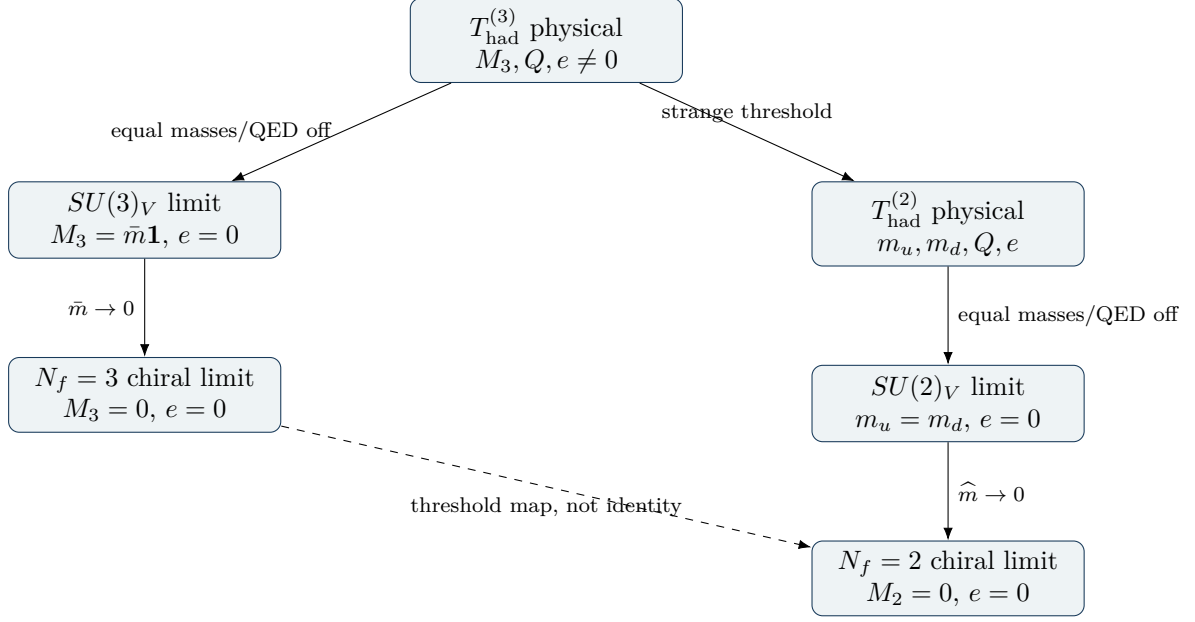


Figure 3: Target-preserving limit arrows and the active-flavor threshold arrow. The latter is not a statement that three generations force three light flavors.

the strange field changes the field content and determinant, so it changes the target fingerprint and requires matching. Generation number indexes UV representation copies and supplies no value for the low-energy active-flavor predicate. $\square$

12 Physical deformations and chiral power counting

12.1 Named counting convention

The declared mesonic counting is

$$p^2 \sim m_q \sim e^2 = \mathcal{O}(\varepsilon^2),$$

with covariant derivatives and external fields counted in the standard chiral scheme. The leading strong Lagrangian is Eq. (8.3). Electromagnetism introduces, at leading order, an invariant of the form

$$e^2 C \text{Tr}(Q_L U Q_R U^\dagger),$$

followed by the renormalized electromagnetic operator basis. This is a grading convention tied to $T_{\chi\text{EFT}}$, not a universal numerical error bound [5, 16].

Table 8: Spurion and power-counting ledger.

Coordinate	Source role	Chiral grade	Primary observable channels
$\hat{m}$ or $\bar{m}$	scalar part of M	$\mathcal{O}(p^2)$	common pseudoscalar masses

Coordinate	Source role	Chiral grade	Primary observable channels
$m_d - m_u$	isovector mass spurion	$\mathcal{O}(p^2)$	strong isospin splittings, mixing
$m_s - \hat{m}$	flavor-octet mass spurion	$\mathcal{O}(p^2)$	$SU(3)$ multiplet splittings
eQ	left/right charge spurions	$e = \mathcal{O}(p)$	electromagnetic splittings, currents
$e^2(m_d - m_u)$	mixed insertion	$\mathcal{O}(p^4)$	mixed strong/EM channels
loop/LEC scale	renormalized EFT data	order dependent	scale cancellation in observables

Theorem 12.1 (SYM6-P04: spurion and power-counting theorem). *The physical isospin and hadronic- $SU(3)$ records possess a coordinate-resolved expansion in the mass, electromagnetic, mixed, derivative, and loop grades of Table 8. At a fixed declared truncation, every omitted term belongs to a typed remainder class. Scheme or operator-basis changes preserve the matched observable when coefficients and loops are transported consistently.*

Proof. Formal source covariance enumerates invariant operators order by order. Freezing the sources identifies explicit deformations without changing their transformation ancestry. Chiral dimension is additive under products and loop counting, so omitted operators have a declared minimum grade. Renormalization moves scale dependence between loops and low-energy constants; the physical amplitude is invariant to the calculated order. No numerical error follows without coefficient and kinematic information. $\square$

12.2 Normal-form deformation fields

The record ISO-PHYS stores $\Delta = (m_d - m_u, e^2, \text{mixed})$, its two-flavor EFT window, matching scheme, and selected invariant observables. HAD3-PHYS stores $\Delta = (m_s - \hat{m}, m_d - m_u, e^2, \text{mixed})$ with three-flavor power counting. The records are not duplicates: they have different field content, convergence envelopes, and target maps.

13 Approximation poset and sector closure

13.1 Two orders with different domains

Limit idealization and EFT accuracy are not one relation. For a common ambient comparison fiber F fixing active fields, algebra, decoupling history, and comparison coordinates, define

$$\sigma_F^{\lim}(a) = (F, Z_a, \tau_a), \quad a \preceq_F^{\lim} b \iff F_a = F_b \text{ and } Z_a \subseteq Z_b.$$

Thus b is at least as constrained or idealized as a , although the two loci have different exact fingerprints τ_a and τ_b . The two-flavor fiber has coordinate universe $\{m_d - m_u, e\}$; the three-flavor fiber has $\{m_s - \hat{m}, m_d - m_u, e\}$. No comparison is formed across those fibers.

Separately, on one exact target T and observable class O , define

$$\sigma_{T,O}^{\text{EFT}}(a) = (T, O, D_a, R_a, S_a),$$

where D_a is the retained operator-grade downset, R_a the typed remainder class, and S_a the registered scheme-equivalence class. EFT comparison is literal componentwise refinement: target and observable class agree, $D_a \subseteq D_b$, the remainder class refines in the registered remainder poset, and $S_a = S_b$. A projection is a morphism of $\mathbb{C}_6$, not an alternative clause in either order.

Theorem 13.1 (SYM6-P05: target-indexed approximation partial order). *The relations $\preceq_F^{\text{lim}}$ and $\preceq_{T,O}^{\text{EFT}}$ are partial orders on their respective declared domains. The ledger derives the limit relation from zero-coordinate inclusion rather than accepting a handwritten pair list. Isospin and three-flavor idealization remain incomparable across fibers, and threshold decoupling remains a morphism rather than an approximation edge.*

Proof. For the limit order, reflexivity and transitivity are those of set inclusion inside fixed F . Mutual inclusion gives $Z_a = Z_b$; together with the registered one-locus-per-signature rule in F , this gives the same normalized locus and proves antisymmetry. Before forming the relation, the audit cross-resolves every locus: the fiber target must have the declared active fields and operator algebra; the locus fingerprint must belong to that target; and the locus record must carry the same target, fingerprint, and action algebra in its own $\text{NF}_6.T$ and $\text{NF}_6.G$. Its EFT signature must resolve back to that same record target. For the EFT order, reflexivity and transitivity follow from operator-grade and remainder refinement, while mutual refinement gives equal downsets and remainder classes; T , O , and S are already fixed. Both proofs are direct and independent of R03. The executable ledger derives exactly $\text{ISO-PHYS} \preceq \text{ISO-STRONG}$ in the two-flavor fiber and $\text{HAD3-PHYS} \preceq \text{HAD3-STRONG}$ in the three-flavor fiber, and rejects cross-fiber comparisons. $\square$

Table 9: Selected approximation comparisons.

a	b	Relation	Reason
physical two-flavor	exact isospin locus	$a \preceq_F^{\text{lim}} b$	isovector mass and QED coordinates set to zero
physical three-flavor	exact vector $SU(3)$ locus	$a \preceq_F^{\text{lim}} b$	octet/isovector mass and QED coordinates set to zero
isospin limit	vector $SU(3)$ limit	incomparable	different target fibers and active fields
two-flavor ChPT order	three-flavor ChPT order	incomparable	different targets and observable classes

13.2 Four-record closure

Table 10: Approximate-flavor normal forms.

Record	Exact target/limit	Deformation and EFT residue
ISO-STRONG	$T_{\text{had}}^{(2)}$, $m_u = m_d$, QED/EW off	exact vector action and currents; no approximation label inside its own target
ISO-PHYS	physical two-flavor hadronic target	$m_d - m_u$, e^2 , mixed terms; two-flavor ChPT envelope
HAD3-STRONG	$T_{\text{had}}^{(3)}$, equal light masses, QED/EW off	exact vector $SU(3)$ action; common nonzero mass allowed
HAD3-PHYS	physical three-flavor hadronic target	strange-light, isovector, EM, mixed grades; three-flavor envelope

Approximate-flavor exit certificate

Four additional atomic records are admitted. Each exact statement is indexed to a decoupled strong-sector target. Every physical approximation carries explicit deformation, scale, scheme, field-content, and remainder coordinates. The poset is proved only inside declared fibers and contains registered incomparabilities. No step derives N_f from generation cardinality or calls isospin exact in the full Standard Model.

Part IV

Custodial symmetry and Higgs handoff

14 Scalar and gauge-Higgs custodial construction

14.1 The fixed Higgs carrier and its matrix representative

Let H be the already fixed Standard Model Higgs doublet with hypercharge 1/2 and define

$$\tilde{H} = i\sigma^2 H^*, \quad \Phi = (\tilde{H}, H).$$

This is an invertible repackaging of the four real components of one carrier, not a second scalar field. It obeys

$$\Phi \mapsto L\Phi R^\dagger, \quad L \in SU(2)_L, \quad R \in SU(2)_R,$$

and $\Phi^\dagger \Phi = (H^\dagger H)\mathbf{1}_2$. The one-doublet potential

$$V(H) = -m^2 H^\dagger H + \lambda (H^\dagger H)^2 = -\frac{m^2}{2} \text{Tr}(\Phi^\dagger \Phi) + \frac{\lambda}{4} [\text{Tr}(\Phi^\dagger \Phi)]^2$$

is therefore invariant under the presentation $SU(2)_L \times SU(2)_R$.

Theorem 14.1 (SYM6-C01: scalar bidoublet and global action). *On $T_{\text{cust}}^{(0)}$, the scalar kinetic term with ordinary derivative and the one-doublet potential carry the faithful action*

$$G_{\text{scalar}}^{\text{faithful}} = \frac{SU(2)_L \times SU(2)_R}{\{(\mathbf{1}, \mathbf{1}), (-\mathbf{1}, -\mathbf{1})\}} \simeq SO(4)$$

on the four real scalar components. The map $H \leftrightarrow \Phi$ is a representative isomorphism and creates no new carrier.

Proof. The trace invariants and $\text{Tr}(\partial_\mu \Phi^\dagger \partial^\mu \Phi)$ are unchanged under constant L, R . The pair $(-\mathbf{1}, -\mathbf{1})$ acts trivially on Φ , while any other class acts nontrivially on some scalar configuration; Theorem 5.1 gives the quotient. The two columns of Φ are fixed algebraic functions of H , so the map is bijective on the real field components and cannot enlarge the carrier inventory. $\square$

14.2 Gauging and the hypercharge coordinate

Write the matrix covariant derivative schematically as

$$D_\mu \Phi = \partial_\mu \Phi - ig W_\mu^a \frac{\sigma^a}{2} \Phi + ig' B_\mu \Phi \frac{\sigma^3}{2}.$$

The first term gauges the left action. Hypercharge gauges the T_R^3 direction (together with the appropriate commuting fermion charge structure in the full theory) and therefore selects a direction in $SU(2)_R$. When $g' = 0$, the gauge-Higgs sector recovers the global right action; when $g' \neq 0$, the complete $SU(2)_R$ is explicitly deformed.

Theorem 14.2 (SYM6-C02: nested custodial limits). *Custodial statements form the following strict target hierarchy:*

1. $T_{\text{cust}}^{(0)}$: scalar action of C01;
2. $T_{\text{cust}}^{(1)}$: gauge-Higgs action with $g' = 0$;
3. $T_{\text{cust}}^{(2, \min)}$: fermion-coupled minimal-SM branch with $g' = 0$, aligned quark Yukawas, and a lepton qualification caused by the absence of ν_R ;
4. $T_{\text{cust}}^{(2, D)}$: Dirac-neutrino branch with $g' = 0$, $Y_u = Y_d$ and $Y_\nu = Y_e$ in a common flavor basis, modulo the basis-invariant orbit statement.

Scalar invariance does not imply invariance of any later target. The minimal-SM lepton sector has a nontrivial full $SU(2)_R$ action only when its unmatched charged-lepton Yukawa is removed or the target is enlarged.

Proof. Equation (14.2) shows the g' obstruction. For quarks, define $Q_R = (u_R, d_R)^T$. The Yukawa terms combine as $-\bar{Q}_L \Phi Y_Q Q_R + \text{h.c.}$ precisely when the two flavor tensors are the same orbit point, represented by $Y_u = Y_d = Y_Q$ in an aligned basis. With $L_R = (\nu_R, e_R)^T$ the same statement gives

$Y_\nu = Y_e$. The minimal branch lacks ν_R , so a nonzero Y_e has no right-doublet partner and obstructs the full matter action. Each additional condition concerns a new field or interaction coordinate, hence none follows from the scalar potential alone. $\square$

Table 11: Nested custodial targets and exact assumptions.

Target	Exact action	Required coordinates	Principal obstruction outside limit
$T_{\text{cust}}^{(0)}$	scalar $SU(2)_L \times SU(2)_R$	one Higgs doublet, invariant potential	noninvariant scalar extensions
$T_{\text{cust}}^{(1)}$	gauge-Higgs	$g' = 0$	hypercharge direction T_R^3
$T_{\text{cust}}^{(2,\min)}$	quark-completed; lepton-qualified	$g' = 0, Y_u = Y_d; Y_e = 0$ for full lepton action	Yukawa splitting, missing ν_R
$T_{\text{cust}}^{(2,D)}$	quark and lepton completed	$g' = 0, Y_u = Y_d, Y_\nu = Y_e$	tensor misalignment/splitting

15 Vacuum stabilizer and deformation ancestry

15.1 Vacuum action

On the registered electroweak vacuum, choose a gauge representative

$$\langle H \rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v \end{pmatrix}, \quad \langle \Phi \rangle = \frac{v}{\sqrt{2}} \mathbf{1}_2.$$

This statement imports the fixed Higgs vacuum branch from the Higgs arc. Gauge-equivalent representatives are skins; the stabilizer of the matrix witness is invariant.

Theorem 15.1 (SYM6-C03: custodial vacuum-stabilizer theorem). *On an exact custodial target, the stabilizer of $\langle \Phi \rangle = (v/\sqrt{2})\mathbf{1}$ under $L\Phi R^\dagger$ is the diagonal subgroup $SU(2)_V$ with the inherited center quotient. The result distinguishes the gauged left redundancy, the global right action, and the diagonal action on physical response data.*

Proof. For $v \neq 0$, $L\langle \Phi \rangle R^\dagger = \langle \Phi \rangle$ is equivalent to $L = R$. Conversely every diagonal pair stabilizes the witness. The common central pair was already quotiented in C01. In the gauge-Higgs target, left multiplication includes gauge redundancy, but the combined transformation preserving the vacuum acts nontrivially on multiplet relations and physical correlators. The stabilizer is therefore not a claim that a new gauge group survives. $\square$

15.2 Independent deformation sources

Define the tensor decomposition

$$Y_+ = \frac{1}{2}(Y_u + Y_d), \quad Y_- \equiv \frac{1}{2}(Y_u - Y_d),$$

and similarly $Y_{\ell,+}, Y_{\ell,-}$ in the Dirac branch. Under the formal right action, the pair of Yukawa tensors is a spurion multiplet. Freezing $Y_- \neq 0$ selects a custodial direction. The minimal branch adds the occupancy defect Δ_{ν_R} = absent rather than pretending it is a numerical Yukawa splitting.

Table 12: Custodial deformation-source matrix.

Source ID	Target coordinate	Invariant witness	Status and observable channel
CUST-DELTA-g1	$g' / \text{gauged } T_R^3$	g'^2 with fixed embedding	explicit gauge-Higgs deformation; neutral/charged response
CUST-DELTA-YQ	Y_u, Y_d	orbit-invariant singular values/commutators of Y_-	quark mass and loop splitting
CUST-DELTA-YL	Y_ν, Y_e	Dirac-branch $Y_{\ell,-}$ invariants	lepton mass and loop splitting
CUST-DELTA-NR	missing ν_R	representation occupancy predicate	minimal-branch structural obstruction
CUST-DELTA-HD	higher operators	basis-invariant amplitude or custodial classifier	EFT deformation; precision observables
CUST-DELTA-RG	running/mixing	anomalous-dimension transport	scale-dependent coefficients, invariant amplitude fixed

Theorem 15.2 (SYM6-C04: custodial deformation ancestry). *Hypercharge gauging, quark Yukawa splitting, lepton/neutrino structure, missing right-handed-neutrino occupancy, and admitted higher-dimensional operators are distinct custodial-deformation coordinates. They may coexist and mix under RG flow, but no basis choice or coefficient relabeling can identify them unless their invariant source and observable maps agree.*

Proof. The sources occupy different target fields and representations: a gauge coupling/embedding, flavor tensors, a field-occupancy predicate, and EFT operators. Their independent removal produces different stabilizers, so F03 forbids collapse. SYM-5 orbit discipline replaces basis entries by invariants, while the RG transport import moves coefficients with their operator basis so invariant amplitudes are preserved. Coincident observable contributions do not imply identical ancestry. $\square$

15.3 Nested-target diagram

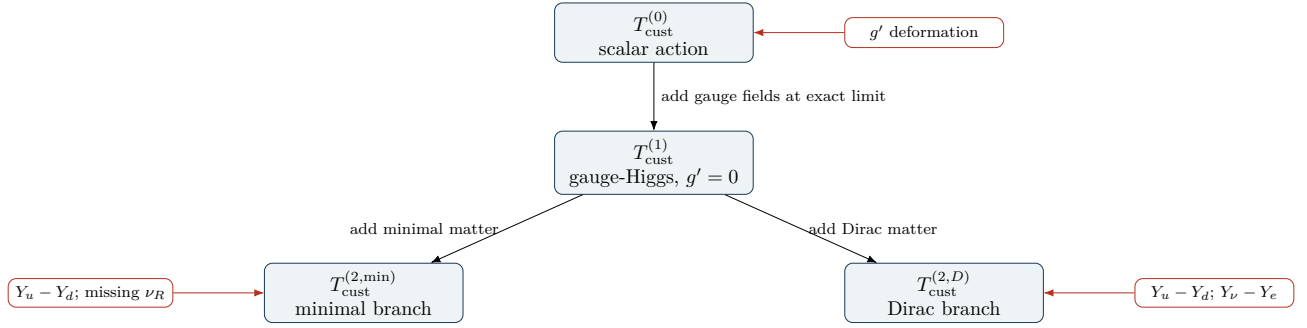


Figure 4: Custodial targets are nested only under explicit assumption vectors. Red arrows are deformation ancestry, not target inclusions.

16 Diagnostics, invariants, and SYM-8 export

16.1 Diagnostic hierarchy

At tree level define

$$\rho = \frac{M_W^2}{M_Z^2 \cos^2 \theta_W}.$$

The one-doublet renormalizable Standard Model gives $\rho = 1$ at tree level. The oblique parameter T measures a selected difference of gauge-boson self-energies [17]. These are consequences or probes in specified approximation classes. They are not the group action itself.

Table 13: Diagnostic implication and failed converse ledger.

Diagnostic	Valid implication	Invalid converse	Counterexample class
exact full target	custodial multiplet relations	none used	definition is the action/stabilizer certificate
tree $\rho = 1$	doublet vacuum gives protected tree relation	$\rho = 1 \Rightarrow$ full custodial action	unequal Yukawas and $g' \neq 0$ coexist with tree relation
$T = 0$ in a fit	selected oblique difference vanishes	$T = 0 \Rightarrow$ no custodial violation	cancellations or nonoblique effects
operator label	may signal violation in one basis	label is invariant classifier	EOM/basis transformation redistributes operators

Diagnostic	Valid implication	Invalid converse	Counterexample class
$\mathcal{T}$ -type invariant	detects declared tree-level violation class	universal all-orders biconditional	scope limited by construction assumptions

Theorem 16.1 (SYM6-C05: custodial diagnostic invariance). *The defining certificate for custodial symmetry is the target-indexed group action, vacuum stabilizer, and deformation ledger. ρ , T , and basis-specific operator labels are diagnostic maps with the one-way implications and counterexamples of Table 13. A reparameterization-invariant combination such as the tree-level $\mathcal{T}$ construction of Kribs et al. [10] is stronger within its scope but is not promoted to an unqualified all-order definition.*

Proof. C01–C04 define the symmetry without observables. The tree relation $\rho = 1$ persists in the one-doublet model although hypercharge and realistic Yukawas deform the full global action, disproving the converse. Oblique T ignores general nonoblique corrections and can vanish by cancellation. Operator bases are related by equations of motion and field redefinitions, so labels are not invariants. The stated invariant constructions remedy specified failures but retain their target and order assumptions. $\square$

16.2 Custodial packet

The SYM-8 export contains:

1. target IDs $T_{\text{cust}}^{(0)}, T_{\text{cust}}^{(1)}, T_{\text{cust}}^{(2,\text{min})}, T_{\text{cust}}^{(2,D)}$;
2. the bidoublet action and faithful quotient;
3. the vacuum stabilizer and gauge/global distinction;
4. the exact assumption vector for each nested limit;
5. independent deformation sources with basis-invariant witnesses;
6. diagnostic implication directions and invalid converses;
7. fixed Higgs-carrier provenance and a prohibition on carrier redefinition.

Theorem 16.2 (SYM6-C06: custodial interface closure). *The atomic records CUST-SCALAR, CUST-GH, CUST-MATTER-MIN, and CUST-MATTER-DIRAC, together with the non-atomic CUST-DIAG-CERT, jointly form one compatible versioned custodial packet. Every consumer field required by the declared SYM-8 schema is populated, and no field alters the identity, occupancy, or primitive role of the imported Higgs carrier.*

Proof. C01 supplies the scalar action and quotient; C02 constructs two separate matter targets rather than a compound branch record; C03 supplies the vacuum stabilizer; C04 supplies each branch’s deformation ancestry; C05 supplies the target-indexed diagnostic certificate and failed converses. Their exact values agree across the target, record, diagnostic, edge, and export registries. Since Φ is an invertible representative of H and every atomic record imports the same Higgs carrier ID, no additional carrier is introduced. $\square$

16.3 Four-record closure

Table 14: Custodial normal-form completion.

Record	Target	Decisive content
CUST-SCALAR	$T_{\text{cust}}^{(0)}$	bidoublet representative, $SO(4)$ faithful action, fixed carrier
CUST-GH	$T_{\text{cust}}^{(1)}$	$g' = 0$ limit, diagonal stabilizer, hypercharge deformation
CUST-MATTER-MIN	$T_{\text{cust}}^{(2,\text{min})}$	quark alignment, missing- ν_R occupancy, minimal-lepton qualification
CUST-MATTER-DIRAC	$T_{\text{cust}}^{(2,D)}$	quark and lepton right-doublet alignment on the Dirac branch

Custodial exit certificate

Four additional atomic records are admitted. Scalar, gauge-Higgs, minimal-matter, and Dirac-matter claims are not conflated. Hypercharge, quark Yukawa splitting, lepton structure, missing ν_R , and EFT operators remain distinct causes. The non-atomic diagnostic certificate is implication-typed and target indexed. The frozen packet can be consumed by SYM-8 without changing the Higgs carrier or reopening a SYM-6 definition.

Part V

Heavy-quark emergence and matching

17 From QCD heavy fields to HQET

17.1 Fixed-velocity field split

Let Q be a heavy quark of mass m_Q and choose a timelike unit velocity $v^2 = 1$. Decompose a momentum in the declared patch as

$$p_Q = m_Q v + k, \quad |k| \ll m_Q,$$

and define $P_{\pm} = (1 \pm \not{v})/2$ together with

$$h_v(x) = e^{im_Q v \cdot x} P_+ Q(x), \quad H_v(x) = e^{im_Q v \cdot x} P_- Q(x), \quad Q = e^{-im_Q v \cdot x} (h_v + H_v). \quad (12)$$

The split is invertible once v and m_Q are fixed. It is not an exact global statement that every QCD mode is soft.

Theorem 17.1 (SYM6-H01: heavy-mode decomposition). *On the velocity patch of $T_{\text{HQET}}^{(N_h, v)}$, Eq. (12) yields $\not{v}h_v = h_v$, $\not{v}H_v = -H_v$ and converts the heavy-quark QCD Lagrangian into*

$$\begin{aligned}\mathcal{L}_Q = & \bar{h}_v i v \cdot D h_v - \bar{H}_v (i v \cdot D + 2m_Q) H_v \\ & + \bar{h}_v i \not{D}_\perp H_v + \bar{H}_v i \not{D}_\perp h_v,\end{aligned}\tag{13}$$

where $D_\perp^\mu = D^\mu - v^\mu v \cdot D$.

Proof. Projector algebra gives the eigenvalue identities and $P_+ P_- = 0$. Substitution into $\bar{Q}(i\not{D} - m_Q)Q$, differentiating the phase, and resolving $\not{D} = \not{v}(v \cdot D) + \not{D}_\perp$ gives Eq. (13). The map is invertible on the chosen velocity patch by the two complementary projectors. $\square$

17.2 Complementary-mode elimination

The H_v equation is

$$(2m_Q + i v \cdot D) H_v = i \not{D}_\perp h_v.$$

Within the low-residual-momentum domain it admits the formal inverse

$$H_v = (2m_Q + i v \cdot D)^{-1} i \not{D}_\perp h_v.$$

Substitution gives the gauge-covariant nonlocal expression

$$\mathcal{L}_{\text{eff}} = \bar{h}_v i v \cdot D h_v + \bar{h}_v i \not{D}_\perp \frac{1}{2m_Q + i v \cdot D} i \not{D}_\perp h_v.\tag{14}$$

Theorem 17.2 (SYM6-H02: HQET expansion and leading action). *Expanding Eq. (14) in $1/m_Q$ and matching hard modes yields*

$$\mathcal{L}_{\text{HQET}} = \bar{h}_v i v \cdot D h_v + \frac{1}{2m_Q} O_{\text{kin}} + \frac{C_{\text{mag}}(\mu)}{2m_Q} O_{\text{mag}} + \mathcal{O}(m_Q^{-2}),\tag{15}$$

with

$$O_{\text{kin}} = \bar{h}_v (i \not{D}_\perp)^2 h_v, \quad O_{\text{mag}} = \frac{g_s}{2} \bar{h}_v \sigma_{\mu\nu} G^{\mu\nu} h_v.$$

The leading action is exact only at order m_Q^0 in the declared fixed-velocity EFT. Coefficients, matrix elements, and the remainder require matching or nonperturbative input.

Proof. Use the geometric expansion of $(2m_Q + i v \cdot D)^{-1}$ and the identity for $(i \not{D}_\perp)^2$ into kinetic and field-strength pieces. Tree matching gives unit coefficient in the displayed normalization; hard loops renormalize the chromomagnetic coefficient and operator basis. The expansion and its domain are standard consequences of the QCD split [11–13, 15]. The omitted terms are typed local operators of dimension six and higher with matching coefficients. $\square$

17.3 EFT-domain certificate

The target stores $(v, N_h, \{m_Q\}, \mu, \text{scheme}, \Lambda_{\text{match}}, n_{1/m})$. Reparameterizations relating nearby decompositions of the same momentum are admitted representative maps only on overlapping patches with coefficient transformations included. A large change of velocity sector is a new target label, not a skin by assertion.

18 Spin/flavor symmetry and current transport

18.1 Leading action

For N_h heavy species, remove each rest mass phase and collect the two projected spin components into

$$\mathbf{h}_v = (h_v^{1,+}, h_v^{1,-}, \dots, h_v^{N_h,+}, h_v^{N_h,-})^T.$$

The leading Lagrangian is

$$\mathcal{L}_0 = \bar{\mathbf{h}}_v i v \cdot D \mathbf{h}_v,$$

which is proportional to the identity in heavy spin/flavor space. The common heavy-number phase is recorded separately from the non-Abelian action; on a fixed-number physical algebra its center kernel is quotiented according to F02.

Theorem 18.1 (SYM6-H03: heavy spin, flavor, and spin-flavor). *At common fixed v and leading order, one heavy species has an $SU(2)_{\text{spin}}$ action, N_h species have a heavy-flavor action, and together the non-Abelian transformations act as $SU(2N_h)$ on $\mathbf{h}_v$, with the heavy-number $U(1)_H$ and any fixed-number kernel recorded separately. The action is faithful only after quotienting the kernel on the declared operator/state algebra. It is not an exact finite-mass QCD symmetry.*

Proof. The gluon coupling $v \cdot A$ contains neither Dirac spin matrices nor a heavy-flavor tensor, so every constant unitary rotation of the $2N_h$ components preserves $\mathcal{L}_0$. Splitting the common phase gives $SU(2N_h)$ and $U(1)_H$ data. Heavy-light interpolating operators transform nontrivially under the non-Abelian action, while central elements that act only as an already fixed number phase lie in the physical-algebra kernel. The kinetic and chromomagnetic terms in Eq. (15) reintroduce mass and spin dependence, proving the leading-limit qualification [14]. $\square$

18.2 Selected current matching

For a heavy-light current,

$$\bar{q} \Gamma Q = e^{-im_Q v \cdot x} \left[\sum_i C_i(m_Q, \mu) \bar{q} \Gamma_i h_v + \frac{1}{2m_Q} \sum_j B_j(m_Q, \mu) O_j + \dots \right]. \quad (16)$$

For heavy-to-heavy currents, source and target velocities and the recoil $w = v \cdot v'$ are explicit. Matching means equality of a declared set of renormalized amplitudes or Green functions to a stated order, not equality of field symbols.

Theorem 18.2 (SYM6-H04: heavy-current matching and RG transport). *For every selected current in the record registry there is a matching matrix $C(\mu)$ and operator vector $O(\mu)$ such that declared QCD and HQET amplitudes agree through the registered order. Scale independence of the matched current gives*

$$\mu \frac{d}{d\mu} C(\mu) = -C(\mu) \gamma_O(\mu),$$

up to the chosen row/column convention. Operator-basis changes act by inverse transformations on coefficients and leave the matched amplitude invariant.

Proof. The EFT operator basis spans all local structures allowed at the declared dimension and quantum numbers. Equating renormalized amplitudes at the hard scale fixes coefficients order by order. Differentiating the scale-independent full-theory current and using $\mu dO/d\mu = \gamma_O O$ gives the RG equation. If $O' = RO$, then $C' = CR^{-1}$, so $C'O' = CO$. Thus basis and scale transport change representatives without generating a new symmetry or deleting a physical correction. $\square$

19 Corrections, observables, and target boundary

19.1 Operator-level correction grades

Table 15: HQET operator and symmetry-grade ledger.

Contribution	Parametric order	Leading symmetry effect	Observable residue
$\bar{h}_v i v \cdot D h_v$	m_Q^0	preserves spin and common-velocity flavor	universal leading multiplet structure
$O_{\text{kin}}/2m_Q$	$1/m_Q$	spin preserving; flavor deformed by $1/m_Q$	spin-averaged mass/kinematic corrections
$C_{\text{mag}} O_{\text{mag}}/2m_Q$	α_s^n/m_Q	breaks heavy spin and combined spin-flavor	hyperfine splitting channels
current coefficients	α_s^n and $1/m_Q$	preserve or deform by operator representation	form-factor normalization/evolution

Contribution	Parametric order	Leading symmetry effect	Observable residue
dimension-six operators	$1/m_Q^2$	operator-specific	Darwin, spin-orbit, time-ordered residues
remainder	$\mathcal{O}((\Lambda/m_Q)^{n+1})$ in power counting	unresolved above truncation	explicit uncertainty class, not a number

Theorem 19.1 (SYM6-H05: heavy-symmetry correction hierarchy). *Every retained HQET correction is graded by $1/m_Q$, perturbative matching, operator representation, and the leading symmetry coordinates it preserves or deforms. O_{kin} preserves heavy spin but carries species-dependent $1/m_Q$ coefficients; O_{mag} breaks heavy spin. The phrase “broken at $1/m_Q$ ” is incomplete without this operator incidence.*

Proof. Heavy-spin rotations commute with $D_\perp^2$ but not with $\sigma_{\mu\nu}G^{\mu\nu}$. Coefficients $1/m_Q$ differ among species, so the flavor rotation mixing unequal masses is deformed even for a spin scalar. Higher operators are tested by their commutator with the leading charges. Additivity of the operator expansion and RG grading produces the correction poset in the machine record. $\square$

19.2 Observable statements and their ceiling

The leading symmetry organizes degeneracies and relates matrix elements. For example, heavy-spin partners become degenerate at leading order and hyperfine splittings begin with the chromomagnetic coordinate. Heavy-to-heavy form factors reduce to universal structures in the symmetry limit, with normalization conditions at special recoil points where the current and state hypotheses apply [14, 18]. SYM-6 registers these invariant patterns, not numerical form factors, decay rates, or matrix elements.

19.3 Heavy-hadron occupancy predicate

Define

$$\Omega_{\text{had}}(Q) = (m_Q \gg \Lambda_{\text{QCD}}) \wedge (|k| \ll m_Q) \wedge (\Gamma_Q \ll \Lambda_{\text{QCD}}) \wedge \text{bound-state target exists.}$$

The last two clauses separate a formal large mass from actual heavy-hadron occupancy. The conventional scale/lifetime input places bottom, and with a less precise expansion charm, inside the declared hadronic family. The top width is not small on the hadronization scale, so the corresponding long-lived top-hadron target is not occupied.

Theorem 19.2 (SYM6-H06: heavy-hadron domain and top exclusion). *The hadronic HQET record set is populated only by flavors satisfying Ω_{had} . In the physical Standard Model, the declared set is $\{c, b\}$ with flavor-dependent control quality. The top quark fails the lifetime/hadronization conjunct even though $m_t \gg \Lambda_{\text{QCD}}$; therefore it is not an occupant of the same heavy-hadron symmetry record.*

Proof. HQET as a short-distance expansion can be written for a sufficiently heavy field, but the heavy-hadron symmetry records also refer to bound states containing that field. Formation requires a QCD timescale of order $1/\Lambda_{\text{QCD}}$, while decay occurs on $1/\Gamma_Q$. The registered physical ordering is $\Gamma_{c,b} \ll \Lambda_{\text{QCD}}$ and $\Gamma_t \gtrsim \Lambda_{\text{QCD}}$, with top decaying before ordinary hadronization. Hence mass alone satisfies only one conjunct and cannot establish target membership. $\square$

19.4 Four-record closure

Table 16: HQET normal-form completion.

Record	Exact envelope	Decisive content
HQ-SPIN	one heavy flavor, common fixed v , leading order	$SU(2)$ action; chromomagnetic correction channel
HQ-FLAVOR	N_h species, common fixed v , leading order	mass-independent leading action after phase removal; finite-mass coefficients deform
HQ-SPINFLAVOR	combined leading target	$SU(2N_h)$ -type faithful action plus separate heavy number
HQ-CORR	matched finite-order HQET	kinetic, chromomagnetic, current, RG, remainder, observables; c, b occupancy

HQET exit certificate

Four final sector records are admitted. The field split and exact intermediate Lagrangian precede the EFT expansion. Leading spin/flavor symmetry is restricted to a common fixed-velocity target and receives a faithful-action qualification. Every finite-mass effect is operator graded. Top is excluded by the occupancy predicate rather than convention. No Wilson coefficient or hadronic form factor is structurally inferred.

Part VI

Integrated ancestry and terminal closure

20 Common UV-to-IR transport graph

20.1 Typed edge grammar

Let $\mathfrak{G}_6 = (V_6, E_6)$ be the directed graph whose nodes are target objects, physical constructions, normalized records, observables, and handoff packets. An edge has the form

$$e = (id, src, \tau_{src}, dst, \tau_{dst}, type, \mu, w, p, q),$$

with exact endpoint fingerprints, registered map μ , typed warrant w , preserved-data IDs p , and assumption IDs q . The permitted generator types are the thirteen types fixed in F01. The free category $\mathbf{C}_6 = \text{FreeCat}(\mathfrak{G}_6)$ adds only empty identities and composition when both the intervening node and its fingerprint agree. A path is complete only if it starts at T_{SM} or a registered conventional source, enters the atomic record, and ends at both a registered observable and a declared handoff or terminal marker.

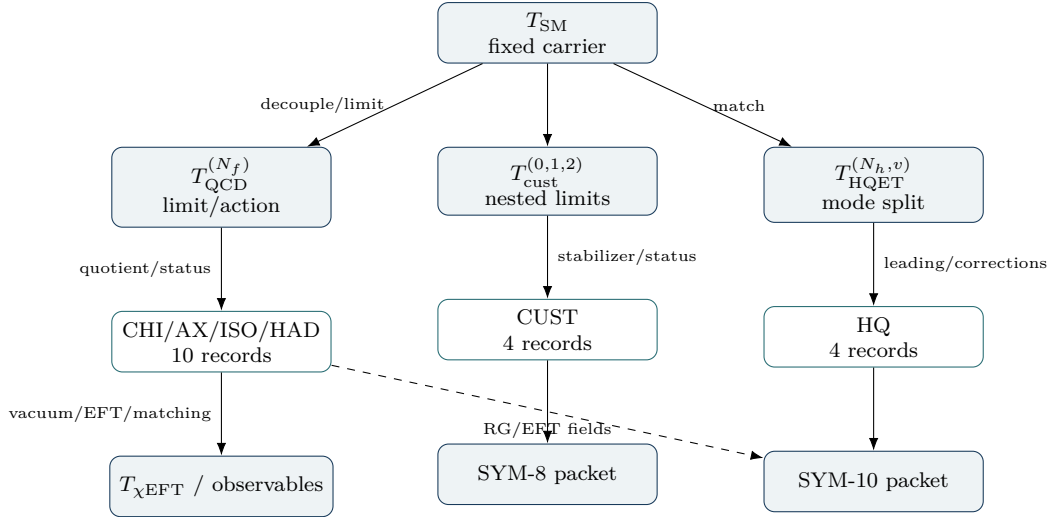


Figure 5: Compressed integrated ancestry graph. The release ledger expands this into one source-to-handoff path for each atomic record.

Table 17: Ancestry path families.

Atomic family	Source path	Terminal residue
CHI	$T_{\text{SM}} \rightarrow T_{\text{QCD}} \rightarrow \text{action} \rightarrow$ faithful quotient $\rightarrow$ currents	exact-limit and vacuum/EFT packets

Atomic family	Source path	Terminal residue
AX	$T_{\text{QCD}} \rightarrow$ classical axial action $\rightarrow$ ABJ/discrete/background split	WZW/Skyrmion anomaly handoff
ISO/HAD3	$T_{\text{QCD}} \rightarrow T_{\text{had}}^{(2,3)} \rightarrow$ stabilizer $\rightarrow$ spurion grades	approximation and observable packet
CUST	Higgs carrier $\rightarrow \Phi \rightarrow$ nested target $\rightarrow$ vacuum/deformations	SYM-8 packet
HQ	QCD heavy field $\rightarrow (h_v, H_v) \rightarrow$ HQET $\rightarrow$ leading action/corrections	SYM-10 packet and observable hierarchy

Theorem 20.1 (SYM6-R01: UV-to-IR ancestry-path construction). *Every one of the eighteen atomic records has at least one complete typed path in $\mathcal{C}_6$. Every generating edge is supported by a named SYM-6 result, a registered import, or an explicitly typed physical input; no path uses notation equality as transport. The $N_f = 3 \rightarrow 2$ change is a decoupling/matching generator, never a restriction.*

Proof. The sector completion tables give the entry theorem for every atomic ID. The edge ledger expands the five path families in Table 17. Before path search, every map and every edge is checked independently against the warrant result’s explicit source-fingerprint, target-fingerprint, record-target-class, family, and edge-type scopes. Each row must then agree exactly with its map entry on source, source fingerprint, destination, destination fingerprint, generator type, warrant, preserved data, and assumptions. If the destination is a record, its own $\text{NF}_6.T$ target class and fingerprint are checked directly as a second target witness. Only validated rows enter the graph. In particular E009 uses V02 for the vacuum-realization construction; V01 remains the registered witness and is not misused as an EFT arrow. The physical justification of each generator is the cited sector theorem; free-category reachability alone is not substituted for it. $\square$

21 Existence, normalization, and multi-cause compatibility

21.1 Record assembly

For each atomic slot, assembly selects the target and invariant IDs, copies theorem-proved fields, retains every imported limitation, sorts set-valued causes, and attaches the certificate. Assembly never fills a physics field from its label. For example, **CHI-VAC-DIAG** receives its vacuum field from V01 and its stabilizer from V02; it is not populated because its name contains “VAC.”

Theorem 21.1 (SYM6-R02: integrated normal-form existence). *Every declared atomic slot has a populated, compatible NF_6 and Cert_6 . All eighteen records are admitted; no required field is blank, unknown, or supported only by a record name.*

Proof. F01–F04 supply the grammar. Q01–M02 close six chiral/axial slots; P01–P05 close four approximate-flavor slots; C01–C06 close the scalar, gauge-Higgs, minimal-matter, and Dirac-matter slots; H01–H06 close four HQ slots. The serialized atlas stores typed target, action, current, anomaly, witness, cause, EFT, matching, observable, theorem, packet, and certificate IDs. Each ID-bearing path has one resolver namespace and a compatibility rule. Result, provenance, import, and assumption entries declare target/family scopes; imports and their provenance entries must agree, and each record must fall within every consumed scope. A recomputed digest cannot rescue an unresolved or wrong-scope value. Each record stores both a full representative-content digest and an invariant-projection digest. Thus every slot has a constructed witness rather than a label-derived value. $\square$

21.2 Canonicalization and collision tests

Define the canonical projection

$$c : \text{Rep}_6 \longrightarrow \text{Inv}_6.$$

It performs:

1. target fingerprint resolution;
2. presentation group replacement by its registered faithful-action class;
3. basis/orbit, current-improvement, and operator-basis normalization through registered transports;
4. sorted anomaly, deformation, and dependency sets;
5. replacement of a raw scheme label by its registered scheme-equivalence class, while retaining envelope, order, and branch;
6. canonical observable and handoff IDs.

The full serialization and invariant projection are different objects and receive different digests. Raw presentation, raw scheme, current-improvement representative, operator-basis label, and chart convention are absent from $c(r)$ after resolution. A difference in target, faithful action, kernel, anomaly, vacuum, branch, EFT order, invariant observable, or export payload remains in $c(r)$ and prevents collision.

Theorem 21.2 (SYM6-R03: normalized classification uniqueness). *For every declared representative r , $c(c(r)) = c(r)$. Moreover,*

$$r \sim_6 r' \implies c(r) = c(r'), \quad c(r) = c(r') \implies r \sim_6 r'$$

within the declared record system. Thus invariantly equal representatives canonicalize to one record, while an invariant difference produces distinct records. Confluence is factorization through c , not an assumption that arbitrary syntactic rewrites commute.

Proof. The invariant tuple consists of the resolved target and branch fingerprint; physical algebra; faithful action, kernel, and image; limit class; current and charge classes; anomaly/vacuum/boundary IDs; cause set; EFT envelope, order, remainder, scheme class, and operator-basis class; matching-map class; invariant observables; exact handoff; and status vector. Resolution and deterministic set sorting leave an already invariant tuple unchanged, proving idempotence. Four positive fixtures change, respectively, a faithful presentation, a consistently transported scheme, a current improvement, and an HQET operator basis, and each obtains the same projection. Every fixture is bound to the transport declared by the registry entry for its changed field, with the exact declared source and target representative IDs. Negative fixtures change target, branch, action, anomaly, vacuum, order, observable, or packet and are rejected or project differently. Equality of canonical tuples is therefore equivalent to $\sim_6$ on the declared representatives. The validator also checks both forbidden collision directions. This is uniqueness of normalized classification, not uniqueness of physical cause. $\square$

21.3 Cause-coordinate incidence

Table 18: Multi-cause incidence matrix. A check means the coordinate may be populated independently.

Record class	anomaly	vacuum	mass	QED/ g'	EFT	RG
CHI/AX	✓	✓	✓	✓	✓	✓
ISO/HAD3	–	branch input	✓	✓	✓	✓
CUST	–	✓	Yukawa	✓	✓	✓
HQ	–	hadron state	$1/m_Q$	–	✓	✓

Theorem 21.3 (SYM6-R04: multi-cause compatibility and no double counting). *Anomaly, explicit deformation, vacuum realization, threshold matching, EFT truncation, and RG transport occupy independently typed fields. Multiple causes may coexist in one record and are counted once each by stable source ID. Two causes are merged only if their source, target action, invariant witness, and observable map are all identical.*

Proof. F03 supplies independent witnesses. The sector theorems type each cause before assembly. The canonical record stores causes as a set keyed by stable source ID, while the cause registry rejects two labels with the same kind, source, and witness signature. This eliminates duplicate presentations while retaining distinct physical origins. A shared observable channel is insufficient for merger because maps with different sources can interfere or cancel. Thus “the symmetry is broken” is replaced by a recoverable incidence vector. $\square$

22 Occupancy exhaustion and no-promotion closure

22.1 Finite candidate decision

The declared candidate generator is the crosswalk from the six top-level families to the eighteen fixed atomic burdens. It is not the class of all imaginable generalized or BSM symmetries. Each burden is decided by its completion theorem; rivals that change target or collapse fields are rejected but do not create undeclared atomic slots.

Table 19: Family occupancy and decision count.

Family	Slots	Admitted	Coverage witness
χ QCD	3	3	Q01–Q04, V01–V03
axial	3	3	A01–A03, M01–M02
isospin	2	2	P01, P03–P05
hadronic $SU(3)$	2	2	P02–P05
custodial	4	4	C01–C06
heavy quark	4	4	H01–H06
Total	18	18	zero undecided declared slots

Theorem 22.1 (SYM6–R05: declared-record coverage and exhaustion). *Every slot in the fixed eighteen-record inventory is decided and admitted, every admitted representative is normalized, and no declared slot remains unclassified. Therefore the SYM-6 inventory is exhausted exactly over $\mathcal{R}_6$ as refined in Table 2. The theorem makes no claim of exhaustive BSM, defect, noninvertible, finite-temperature, or all-IR-symmetry classification.*

Proof. Table 19 is a disjoint partition by stable record ID. R02 proves existence for each row and R03 proves one normalized record per invariant slot. The candidate register and atlas have the same eighteen IDs, verified by set equality in the executable audit. The stated exclusions are outside the generator, so they neither create missing declared slots nor become silently rejected universal candidates. $\square$

22.2 Primitive-role comparison

Table 20: Family-by-family no-promotion certificate.

Family	Upstream support used	Why effective standing is not a new primitive carrier
χ QCD/axial	color/matter carriers, anomaly and vacuum interfaces	limit action and IR realization depend on fixed QCD fields, gauge action, current, and matching data
ISO/HAD3	same QCD carriers plus mass/QED coordinates	stabilizers of selected parameter limits; no independent bearer outside those carriers
CUST	fixed Higgs bridge, gauge/matter carriers, Yukawa/vacuum data	bidoublet is a representative; diagonal action is a limit/vacuum relation, not a new field
HQ	fixed heavy-quark matter and color transport	symmetry arises after mode split and leading EFT projection; finite-mass corrections expose dependency

Theorem 22.2 (SYM6-R06: effective-symmetry no-promotion). *No admitted effective, approximate, anomalous, or vacuum-realized SYM-6 record acquires independent primitive Standard Model carrier standing merely because it survives to the IR, organizes observables, defines a conserved limit charge, or appears as an EFT invariance.*

Proof. The upstream carrier closure fixes the same-scope primitive carrier package. Table 20 maps each record family to that package and to the target operations on which it depends. Removing those supports eliminates the record’s field action, target, or transport map. Conversely, no SYM-6 construction introduces an independently occupied field/role needed to preserve Standard Model claim identity: chiral/vector actions are transformations of fixed quarks; custodial Φ is an invertible Higgs representative; HQ fields are projections of a fixed heavy quark. The records are physically real and consequence-bearing, but their role is limit, obstruction, realization, approximation, or transport. A same-target candidate that is not so supported and satisfies the kernel primitive-role burdens would defeat the theorem; none occurs in the exhausted inventory. $\square$

23 Downstream exports and arc synchronization

23.1 Frozen packets

The machine file `sym6_exports.yaml` contains three packets.

1. SYM8-CUSTODIAL-v2: four atomic custodial targets, faithful actions, stabilizer, branch-specific deformation records, non-atomic diagnostic certificate, Higgs-carrier ID.

2. **SYM10-EFT-RG-v2**: chiral/HQET targets, threshold-decoupling maps, scale and scheme fields, derived common-fiber approximation signatures, HQ correction hierarchy, no-generation rule.
3. **CAPSTONE-ANOM-IR-v3**: faithful QCD global form, generic axial image/kernel, exact $N_f = N_c = 3$ left-subgroup map, three-flavor WZW target, completed Skyrmion divergence/inflow, and a no-general- N_f export lock.

Exports contain stable record IDs rather than copied prose. A consumer must either use the exact packet version or register a target update.

23.2 Nonredundancy with SYM-0–5

SYM-6 imports the anomaly, flavor, Higgs, and carrier interfaces but adds the sector constructions that were absent upstream: faithful massless-QCD global action in its declared algebra, registered vacuum-to-coset chain, approximation poset, branch-specific custodial hierarchy, HQET field and correction derivation, and integrated finite record closure. The export diff marks inherited fields, specialized fields, and new fields. Therefore the paper is neither a duplicate survey nor an illicit re-proof of frozen upstream endpoints.

24 Rival ledger, hostile-referee map, and exact nonclaims

24.1 Standing conditions for an objection

A defeating objection must do at least one of the following: break kernel necessity; deny that the stated target or physical object is determinate; invalidate a named import or its specialization; break a sector calculation on its fixed target; construct a same-target counterexample to normalization or role classification; show a missing declared candidate; or break R01–R07. Relabeling the endpoint, attacking a nonclaim, or switching targets without an arrow does not defeat the theorem proved.

The v3.1 bounded review contract is frozen in `FROZEN_REFEREE_CONTRACT.md`. It admits findings only against the five repaired obligations—P05 locus/fiber cross-resolution, R03 transport-bound canonical projection including an operator basis, F04 scoped semantic compatibility, F01/R01 result-scoped maps and destination-record target checks, and the certificate-bound **AX-TH00FT** subgroup map—or a demonstrated patch regression, internal contradiction, or faithful same-target counterexample. Extensions and stronger endpoints are nonblocking.

Table 21: Hostile-referee attack map.

Attack	Load-bearing disposition	Break point if successful
Familiar QFT merely relabeled	New object is integrated normal form, typed ancestry, uniqueness, finite coverage, and no-promotion closure	R01–R07
AASC manufactures QCD dynamics	V01 registers the vacuum witness and preserves its limitation	V01 provenance
Naive chiral direct product	All four denominator identifications and equivalent baryon-normalized quotient are explicit	Q02
Anomaly, mass, and vacuum conflated	Independent equations and status fields	F03, A01–A03, R04
Discrete residue overclaimed	Axial generator is embedded and its graded-local kernel is computed for odd/even N_c	A02
Goldstone conclusion lacks hypotheses	Locality, charge domain, witness and exact limit are explicit	V03
WZW/Skyrmion transport asserted incompletely	The $N_f = N_c = 3$ left-subgroup map, WZW level and variation are fixed; baryon neutrality then forces the completed current, divergence, and inflow	A03, M01–M02
$N_f = 3$ imported from generations	Active-flavor target and threshold map explicit	Q01, P03
Isospin called exact SM symmetry	Exact only on $T_{\text{had}}^{(2)}$	P01
$SU(3)$ approximation universally ranked	Limit order is derived only in a common fiber; EFT accuracy has its own exact-target order	P05
Scalar custodial promoted to full SM	Four nested targets and branch conditions	C02
Hypercharge/Yukawa causes lumped	Independent source matrix	C04
Neutrino branch hidden	Minimal and Dirac targets separately closed	C02, C04
$\rho = 1$ or $T = 0$ definitional	Failed converses and counterexamples explicit	C05
Operator label treated as invariant	Basis-invariant amplitude/classifier required	F02, C05

Attack	Load-bearing disposition	Break point if successful
HQET introduced by slogan	Exact field split and complementary-mode elimination shown	H01, H02
Finite-mass heavy symmetry overclaimed	Leading target and correction operators separated	H03, H05
Top counted as heavy hadron	Occupancy predicate includes lifetime ordering	H06
RG flow generates primitive standing	RG transports a matched record; it does not create its carrier	H04, R06
Unique ancestry means one cause	Normalization and cause incidence are different theorems	R03, R04
Exhaustion claims all IR/BSM symmetries	Candidate generator is the declared eighteen-slot set	R05
No-promotion denies physical reality	The conclusion is role-specific; observables and charges remain real	R06
Exports are not reproducible	Stable IDs, schemas, versioned packets and hashes supplied	R07

24.2 Residual open burdens

The following burdens remain outside the endpoint: dynamical derivation of the QCD condensate and its value; numerical chiral low-energy constants and hadronic spectrum; finite-temperature/density phase structure; all-order global-form and noninvertible-sector completion; numerical precision-electroweak analysis; all BSM custodial classes; nonperturbative HQET coefficients and form factors; and a universal inventory of effective symmetries. A later paper may close one of these only by a target update or a source-identified new theorem, not by rereading R05 as universal.

25 Release certificate and reproducibility package

25.1 Executable audit boundary

The finite audit mechanizes closed schemas and cross-registry semantics: target/branch atomicity; action/kernel resolution; exhaustive semantic namespaces; scoped provenance, result, import and assumption compatibility; independently result-scoped maps and edges with destination-record $NF_6.T$ checks; theorem-DAG acyclicity; full-content and invariant digests; transport-bound representative fixtures; P05 locus/fiber cross-resolution; the digest-bound axial subgroup certificate; both collision directions; path reachability; exact export payloads; frozen mutation failures; and derived release

predicates. Physical QFT propositions remain named theorem assumptions or equation certificates. A passing semantic audit is not represented as proof of the ABJ coefficient, QCD vacuum, WZW variation, or top lifetime ordering.

Theorem 25.1 (SYM6-R07: interface and release closure). *The v3.1.0 release freezes all eighteen normalized records, the fingerprinted ancestry category, the rival ledger, the non-atomic custodial diagnostic certificate, the axial subgroup evidence certificate, and the exact SYM-8/SYM-10/capstone packets. The executable audit derives all twenty release-gate outcomes from predicates. Per-record content and invariant digests fix different identities, the manifest fixes every payload file except itself, and the detached ZIP digest fixes the archive bytes.*

Proof. R05 fixes the record set and R06 its role conclusion. The export registry names exact targets, branches, fingerprints, actions, handoffs, packet locks, and required edges. The validator reconstructs invariant projections, verifies both record digests, checks the theorem DAG and frozen mutation suite, and computes every gate; no input PASS cell is trusted. The build fixes time, locale, PDF metadata, report contents, file ordering, and archive timestamps. Two fresh-extraction builds must reproduce identical PDF, report, manifest, and ZIP bytes. The full-content digest, invariant digest, payload manifest, and detached ZIP digest fix distinct non-self-referential identities. $\square$

25.2 Twenty release gates

Table 22: Release-gate certificate.

No.	Gate	Validated condition	Status
1	target	every record and edge names a target	PASS
2	faithfulness	group records include kernel/faithful action	PASS
3	current	continuous actions include current/action domain	PASS
4	anomaly	ABJ, 't Hooft, global types separated	PASS
5	vacuum	witness input distinguished from consequence	PASS
6	Goldstone	hypotheses, coset, count and EFT map explicit	PASS
7	matching	UV/IR anomaly variation certificate present	PASS
8	approximation	deformation, scale, scheme and order present	PASS
9	custodial	scalar, gauge-Higgs, matter, neutrino branches distinct	PASS
10	diagnostic	invalid ρ/T converses blocked	PASS
11	HQET	split, expansion, leading symmetry, corrections constructed	PASS
12	domain	top excluded by occupancy predicate	PASS
13	normal form	all required eleven fields populated	PASS
14	uniqueness	collisions resolved by $\sim_6$	PASS
15	multi-cause	simultaneous causes retained without duplicates	PASS

No.	Gate	Validated condition	Status
16	coverage	eighteen of eighteen declared slots decided	PASS
17	no-promotion	family role comparison complete	PASS
18	rival	all registered serious rivals disposed	PASS
19	export	SYM-8/SYM-10/capstone schemas validate	PASS
20	release	deterministic report/PDF/manifest/archive identity verified twice	PASS

25.3 Integrated occupancy closure theorem

Theorem 25.2 (SYM-6 integrated occupancy closure). *Let $\mathcal{R}_6$ be the declared six-family domain refined by Table 2. Suppose the registered Standard Model, Higgs-carrier, anomaly, flavor, vacuum, and RG/EFT imports are valid on their stated targets; suppose Q01–H06 hold; and suppose each physical existence witness is used only within its certificate scope. Then:*

1. *every declared candidate is admitted or rejected;*
2. *every admitted candidate has a complete NF_6 and Cert_6 ;*
3. *each record is unique up to $\sim_6$;*
4. *anomaly, vacuum realization, explicit deformation, approximation, and EFT emergence remain distinct compatible coordinates;*
5. *every record has a certified ancestry path and observable/handoff residue;*
6. *the eighteen-slot family is exhausted;*
7. *no admitted effective record acquires independent primitive Standard Model carrier standing;*
8. *the custodial and scale-transport interfaces are frozen for SYM-8 and SYM-10.*

Proof. Items 1–2 are R02 and R05. Item 3 is R03. Item 4 is F03 and R04. Item 5 is R01. Item 6 is R05. Item 7 is R06 under the imported carrier closure and kernel applicability. Item 8 is R07. Every dependency, specialization, and nonclaim is exposed in the theorem and machine registers; no stronger universal or numerical conclusion is used. $\square$

Final endpoint

SYM-6 closes the declared effective-symmetry layer as a constructive, fixed-domain ancestry and occupancy theorem. Its novelty is the synchronized faithful-action, status, vacuum, deformation, EFT, matching, observable, and role record across sectors. Its ceiling is equally exact: registered dynamics remain registered, numerical data remain uncomputed, and the finite inventory is not silently universalized.

A Theorem and dependency inventory

Table 23: Complete forty-result theorem ladder.

ID	Result	Immediate dependencies
SYM6-F01	fingerprinted target-category formation	SYM-0–5 target discipline; fixed SM target
SYM6-F02	faithful-action quotient rule	SYM-1 action/quotient discipline
SYM6-F03	orthogonal status separation	SYM-3 anomaly classes; vacuum/RG imports
SYM6-F04	semantic normal-form admissibility	kernel certificate closure; F01–F03
SYM6-Q01	massless-QCD target/action	fixed QCD carrier; SYM-5 target discipline
SYM6-Q02	faithful chiral group	F02, Q01, global-form data
SYM6-Q03	currents/Ward identities	Q01–Q02, SYM-3 current discipline
SYM6-Q04	mass/EM deformation	Q01–Q03, SYM-5 spurions
SYM6-A01	continuous axial divergence	Q03, SYM-3 Bardeen packet
SYM6-A02	discrete axial residue	F02, A01, SYM-4 discrete ledger
SYM6-A03	anomaly-type separation	F03, SYM-3 complete anomaly engine
SYM6-V01	registered vacuum witness	Q01–Q02, gauge-vacuum import, physical input
SYM6-V02	vacuum stabilizer/coset	Q02, V01, SYM-5 stabilizer discipline
SYM6-V03	Goldstone/nonlinear realization	V02, registered Goldstone hypotheses
SYM6-M01	WZW/anomaly transport	A03, V03, SYM-3 QCD3 WZW packet
SYM6-M02	baryon/Skyrmion residue	M01, declared IR boundary conditions
SYM6-P01	exact isospin limit	Q04, two-flavor hadronic target
SYM6-P02	exact hadronic $SU(3)$ limit	Q04, three-flavor hadronic target
SYM6-P03	flavor-limit refinement	Q01–Q04, P01–P02
SYM6-P04	spurion/power counting	Q04, P03, ChPT sources
SYM6-P05	approximation partial orders	F01, P03–P04, common-fiber signatures
SYM6-C01	scalar bidoublet/action	fixed Higgs carrier, F02
SYM6-C02	nested custodial limits	C01, SYM-4 neutrino branch, SYM-5 Yukawas
SYM6-C03	custodial vacuum stabilizer	C01–C02, Higgs vacuum import
SYM6-C04	custodial deformation ancestry	C02, SYM-5 invariants, RG import
SYM6-C05	custodial diagnostics	C02–C04, invariant source discipline
SYM6-C06	custodial interface closure	C01–C05, fixed Higgs identity
SYM6-H01	heavy-mode decomposition	fixed QCD heavy field, F01
SYM6-H02	HQET expansion/leading action	H01, EFT matching discipline
SYM6-H03	heavy spin/flavor action	F02, H02
SYM6-H04	current matching/RG transport	H01–H03, RG import
SYM6-H05	correction hierarchy	H02–H04

ID	Result	Immediate dependencies
SYM6-H06	heavy-hadron domain/top exclusion	F01, physical lifetime input, H01–H05
SYM6-R01	fingerprinted ancestry path construction	F01 and all sector results
SYM6-R02	normal-form existence	F04, R01
SYM6-R03	canonical-projection uniqueness	F02–F04, R02
SYM6-R04	multi-cause/no double counting	F03, R02–R03
SYM6-R05	declared coverage/exhaustion	R02–R04
SYM6-R06	effective no-promotion	kernel/carrier closure, R05
SYM6-R07	interface/release closure	R05–R06, export schemas

B Imported theorem register

Table 24: Human-readable import digest; machine details are in `sym6_imports.csv`.

Import ID	Imported conclusion	SYM-6 use	Limitation retained
SYM0-ATLAS	status/record grammar	normal-form conventions	no sector-specific physics
SYM1-FAITHFUL	action/global quotient discipline	F02, Q02, C01, H03	target algebra required
SYM2-ANCHOR	Poincare/CPT domain anchor	spacetime background	not reclassified here
ANOM-F-SM-LOCAL-1GEN	SM local anomaly ledger	UV consistency context	one-generation aggregate only
ANOM-F-VA-BARDEEN-DESCENT	vector/axial counterterm discipline	A01–A03	redistribution is not cancellation
ANOM-F-QCD-U1A-NF3	three-flavor singlet axial packet	$N_f = 3$ specialization	not general N_f
ANOM-F-QCD3-WZW-MATCH	three-flavor WZW packet	M01	exact target/level only
SYM4-DISCRETE	discrete/strong- θ ledger	A02 and target boundary	no reopening strong-CP endpoint
SYM4-NU-BRANCH	minimal/Dirac neutrino branch	C02/C04	branch must remain explicit
SYM5-SPURION	basis, orbit, stabilizer, spurion invariants	Q04, P04, C04/C05	QCD chiral group not supplied

Import ID	Imported conclusion	SYM-6 use	Limitation retained
SM-CARRIER-CLOSE	fixed primitive carrier package	R06	same-scope carrier domain
HIGGS-CARRIER	bridge carrier identity	C01–C06	no second Higgs carrier
HIGGS-VACUUM	registered electroweak vacuum branch	C03	empirical/physical branch input
GAUGE-VAC	vacuum-standing discipline	V01, C03	no QCD condensate derivation
RG-TRANSPORT	scale/scheme/matching preservation	P04/P05, C04, H04	transport does not generate symmetry
YUKAWA-SURVIVAL	tensor/spurion survival classes	C02/C04	numerical Yukawas not derived

C Atomic normal-form atlas

The complete machine atlas is `sym6_records.yaml`. Table 25 is its citable digest. “N/A by type” means the schema contains a reasoned inapplicability object, not a missing value.

Table 25: Atomic atlas digest.

Stable record	Target	Faithful action/status	Matching, observable, handoff
CHI-NONSINGLET	$T_{\text{QCD}}^{(N_f)}$	quotient chiral action; nonanomalous local currents	source Ward identities; chiral packet
CHI-VECTOR	$T_{\text{QCD}}^{(N_f)}$	diagonal/vector and $U(1)_B$	baryon/ flavor charges; hadronic limits
CHI-VAC-DIAG	$T_{\chi\text{EFT}}^{(N_f)}$	conditional diagonal stabilizer	Goldstone EFT; anomaly/IR packet
AX-CONT	$T_{\text{QCD}}^{(N_f)}$	classical $U(1)_A$, ABJ obstructed	divergence observable; axial ledger
AX-DISC	QCD graded local algebra	computed odd/even- N_c faithful image	axial embedding and background probes
AX-THOOF	QCD quotient backgrounds	mixed/discrete obstruction	WZW plus completed-Skyrmion matching
ISO-STRONG	$T_{\text{had}}^{(2)}$	exact $SU(2)_V$ limit	isospin currents; approximation packet

Stable record	Target	Faithful action/status	Matching, observable, handoff
ISO-PHYS	physical two-flavor	mass/EM deformed	ChPT observables and errors
HAD3-STRONG	$T_{\text{had}}^{(3)}$	exact vector $SU(3)$	octet currents; approximation packet
HAD3-PHYS	physical three-flavor	strange-light/isovector/EM deformed	ChPT observables and errors
CUST-SCALAR	$T_{\text{cust}}^{(0)}$	$(SU2_L \times SU2_R)/Z_2$	scalar invariants; SYM-8
CUST-GH	$T_{\text{cust}}^{(1)}$	$g' = 0$, diagonal vacuum	gauge-boson relations; SYM-8
CUST-MATTER-MIN	$T_{\text{cust}}^{(2,\text{min})}$	quark alignment / minimal-lepton obstruction	branch invariants; SYM-8
CUST-MATTER-DIRAC	$T_{\text{cust}}^{(2,D)}$	quark and lepton right-doublet alignment	branch invariants; SYM-8
HQ-SPIN	$T_{\text{HQET}}^{(1,v)}$	leading $SU(2)_{\text{spin}}$	multiplet/form-factor relations; SYM-10
HQ-FLAVOR	$T_{\text{HQET}}^{(N_h,v)}$	leading heavy-flavor action	matched currents; SYM-10
HQ-SPINFLAVOR	common leading target	$SU(2N_h)$ -type quotient	unified leading relations; SYM-10
HQ-CORR	finite-order HQET	kinetic, magnetic, and current grades	correction observables; SYM-10

D Schema and formal-audit boundary

D.1 Mechanized statements

The executable audit checks the following finite propositions:

1. every record ID is unique and belongs to the declared inventory;
2. every record has a known target and all eleven normal-form fields;
3. every certificate has provenance, assumptions, witnesses, dependencies, machine data, rival disposition, nonclaims, and release ID, while every record has verified full-content and invariant digests;
4. every ID-bearing record path resolves through exactly one semantic namespace; provenance, results, imports, and assumptions pass target-class, family, and result or edge-type compatibility rules;
5. every result declares source- and target-fingerprint scopes, record-target-class and family scopes,

- and permitted edge types; every map and ancestry edge is checked against both endpoint scopes and, for a record destination, against that record's own $\text{NF}_6.T$;
6. every record is reachable from the fixed source and reaches an observable/handoff through validated edges;
 7. full-content and invariant digests verify, the canonical projection is idempotent on four transport-bound representative fixtures including an operator-basis change, and both collision directions fail closed;
 8. every P05 locus cross-resolves against its fiber target's fields and algebra and its record target; the common-fiber limit and exact-target EFT relations are derived partial orders, with cross-fiber comparisons rejected;
 9. the axial subgroup map cryptographically references a certificate whose action, order, backgrounds, quotient endpoints, evidence, and assumptions agree with the action and background registries;
 10. all eighteen candidates are decided;
 11. all forty result IDs occur once in manuscript source;
 12. all export source IDs and required fields exist;
 13. all twenty release gates are derived from executable predicates and pass;
 14. the frozen legacy and v3.1 mutation families are rejected, including unresolved or wrong-scope semantic IDs, canonical collisions, synchronized wrong-sector edge/map mutations, cyclic theorem/remainder graphs, and axial-certificate corruption.

D.2 Statements deliberately not mechanized as datatype facts

The validator does not prove the functional measure anomaly, the QCD vacuum witness, the Goldstone theorem, WZW variation, the electroweak vacuum, the HQET asymptotic expansion, or physical lifetime orderings. Those appear as named equations, proofs, imports, or registered inputs in the manuscript. This boundary prevents a finite record check from masquerading as a proof of its physical premises.

E Rival and counterexample digest

Table 26: Principal rival dispositions.

Rival claim	Counterexample or type failure	Disposition
naive direct-product chiral group	common flavor/vector and color-center elements act trivially	quotient by generated kernel
mass repairs axial anomaly	mass and measure terms coexist in Eq. (6)	separate Δ and A fields

Rival claim	Counterexample or type failure	Disposition
vacuum breaks $U(1)_A$ instead of anomaly	anomaly exists before vacuum selection	rejected coordinate substitution
WZW follows from coset alone	level depends on UV anomaly/ N_c	require M01 certificate
three generations imply $N_f = 3$	generation and active-flavor targets differ	invalid target map
isospin exact in full SM	QED/weak/Yukawa coordinates present	restrict to $T_{\text{had}}^{(2)}$
scalar custodial means full matter custodial	realistic $Y_u \neq Y_d$; minimal branch lacks ν_R	nested targets
$\rho = 1$ iff custodial	tree one-doublet relation with other breaking sources	only one-way diagnostic
HQ symmetry exact for finite m_Q	O_{mag} and $1/m_Q$ coefficients	leading EFT status
top is a heavy-hadron occupant	decay-before-hadronization ordering	fails Ω_{had}
unique record means unique cause	one record stores a set of causes	R03/R04 distinction
no-promotion makes EFT unreal	role is denied, physical action/observables retained	category error

F Artifact and version contract

Table 27: Canonical release artifacts.

Artifact	Contract
<code>main.tex</code> , <code>sections/*.tex</code>	complete manuscript source; forty result IDs
<code>references.bib</code>	literature and corpus source register
<code>ledgers/sym6_targets.yaml</code>	target branches, fingerprints, and admitted generator types
<code>ledgers/sym6_records.yaml</code>	eighteen typed normal forms, certificates, full-content digests, and invariant digests
<code>ledgers/sym6_imports.csv</code>	theorem-level imports, consumers, limitations
<code>ledgers/sym6_edges.csv</code>	typed, fingerprinted ancestry generators
<code>ledgers/sym6_rivals.csv</code>	rival/counterexample dispositions
<code>ledgers/sym6_exports.yaml</code>	frozen downstream packet schemas

Artifact	Contract
<code>ledgers/sym6_schema.json</code>	machine validation schema
<code>ledgers/sym6_actions.yaml</code> , <code>sym6_causes.yaml</code>	faithful-action/kernel and physical-cause registries
<code>ledgers/sym6_observables.yaml</code>	invariant observable registry
<code>ledgers/sym6_diagnostics.yaml</code>	non-atomic target-indexed diagnostic registry
<code>ledgers/sym6_semantics.yaml</code>	closed current, anomaly, witness, EFT, matching, provenance, assumption, and machine namespaces
<code>ledgers/sym6_maps.yaml</code>	exact object fingerprints, maps, warrants, preserved data, and assumptions
<code>ledgers/sym6_approximation.yaml</code>	common-fiber and exact-target order signatures
<code>ledgers/sym6_representatives.yaml</code>	positive and negative canonical-projection fixtures
<code>ledgers/sym6_axial_subgroup_certificate.yaml</code>	digest-bound subgroup action, order, backgrounds, quotient, and evidence
<code>ledgers/sym6_results.yaml</code>	acyclic forty-result dependency DAG
<code>ledgers/release_gate_spec.yaml</code>	executable gate predicates
<code>ledgers/release_gates.csv</code>	validator-derived gate outcomes
<code>tools/validate_release.py</code>	executable finite audit
<code>ledgers/sym6_validation_report.txt</code>	generated audit result
<code>SHA256SUMS.txt</code>	post-build byte identity for the release files

Release v3.1.0 freezes the stable record IDs, the exact three-flavor anomaly packet and axial subgroup certificate, the scoped semantic registries, and the bounded referee contract. Record content and invariant digests, the payload manifest, and the detached archive digest fix distinct non-self-referential identities. A change of target, sector theorem, record identity, or export payload requires a new version and hashes. Downstream consumers cite the packet ID and release version, not an unversioned filename.

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